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Three-stage optimal scheduling of distribution system with multi-microgrid incorporating wind-solar-hydro-hydrogen-storage☆

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ABSTRACT

With the increasing integration of renewable energy into distribution system, the issue of renewable energy curtailment has become increasingly prominent. Forming a multi-energy complementary and multi-storage coordinated multi-microgrid system is advantageous for the on-site consumption of renewable energy. In this paper, a multi-time scale distributed scheduling strategy is proposed for a multi-microgrid system incorporating wind, solar, hydro, hydrogen and storage, considering source-load uncertainties. Firstly, an energy management framework is established, which achieves global optimization goals while protecting the privacy within microgrids, considering microgrid topology. Secondly, considering the differentiated regulation characteristics of different resources, a scheduling strategy is developed based on rolling optimization, encompassing monthly, weekly, and daily time scales. This strategy effectively utilizes the electric power or energy regulation capabilities of different resources at different time scales. Thirdly, a three-stage electric power and energy balance model is established based on fuzzy chance constraints, and solved by using a variable step-size alternating multiplier method. Finally, a cost allocation method between microgrids and distribution system is achieved based on their respective contributions to the power and energy balance. Case study results demonstrate that the proposed method exhibits good adaptability across different time scales, effectively enhancing the renewable energy absorption capacity.

1. Introduction

1.1. Background and motivation

As environmental pollution and resource scarcity issues intensify, the proportion of distributed energy resources (DERs) in the distribution system (DS) continues to rise [1,2]. Taking Yunnan Province in China, which is abundant in water resources, as an example, during the first half of 2023, the proportion of clean energy generation from wind, solar, and hydro sources reached 78.16%, with hydroelectric station (HS) contributing 65.87% of the total [3]. However, renewable energy generation is subject to fluctuations and uncertainties across multiple time scales due to factors such as climate variability. The escalating problem of curtailment arises from the unpredictability of renewable energy output and suboptimal scheduling practices, posing significant challenges to the secure, economic, and flexible operation of the DS [4,5]. The development of energy storage technologies with distinct temporal characteristics, represented by electrochemical energy storage and hydrogen storage, has significantly facilitated the integration of renewable energy sources across different timescales. As of the end of 2023, the global installed capacity of electrochemical energy storage (EES) systems reached approximately 110 GW [6], and this scale continues to expand driven by the continuous decline in battery costs. Moreover, the green hydrogen will power fuel cell vehicles and hydrogen internal combustion engine systems as sustainable alternatives to conventional gasoline-powered transportation. The proposed transition is projected to enhance overall fuel utilization efficiency while substantially mitigating greenhouse gas emissions across the mobility sector [7]. Hydrogen storage system (HSS) technology as a long-duration energy storage technology has garnered widespread attention due to their characteristics of cleanliness, high efficiency, and convenient transportation [8,9]. At the end of October 2023, globally, operational green hydrogen projects had a cumulative electrolyzer capacity of 1106 MW,

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with projects under construction totaling 14.1 GW [10]. Hydrogen storage plant project is considered effective means to enhance the flexibility of DS over long time scale with possible economic incentives [7]. Therefore, conducting research on multi-time scale coordinated scheduling considering hydrogen storage is crucial for regions with a high proportion of renewable energy sources [11,12].

Moreover, the modern distribution system is characterized by a large number of DERs and market participants, presenting challenges for enhancing system coordination capabilities. To effectively guide the coordinated utilization of DERs, DS are transitioning towards distributed intelligent DS. Microgrids (MGs), as autonomous and controllable entities within the distribution system (DS) [13], can significantly improve the on-site absorption of renewable energy when integrated into MGs [14]. With the rapid development of DERs, MGs and DS are coupled more closely. The coordination of MGs and DS is an innovative way for maximizing synergistic effects when MGs are located in close proximity to DS, because such an coordination can increase the reliability and flexibility of energy supply by allowing the sharing of energy resources among MGs and DS [15]. Interconnecting multiple MGs to form a multi-microgrid (MMG) system further enhances the spatial utilization efficiency of DERs [16]. DERs in MGs and DS can be optimally dispatched by running a coordinated scheduling over the entire system. Developing effective and rapid solutions for the coordination of DS and MGs has become a current research focus. Therefore, with the increasing diversity and volume of demand-side regulation, addressing power and energy balance challenges across larger spatial scales and longer time horizons becomes essential. The coordinated scheduling of DERs in modern DS should be considered from both temporal and spatial perspectives.

1.2. Literature review

Existing studies on coordinated scheduling for DS from temporal and spatial perspective are reviewed.

(1) Multi temporal coordinated scheduling: Renewable energy sources and loads exhibit multi-time scale uncertainties and volatility. The coordination of hydropower, wind and photovoltaics power in DS can alleviate the fluctuation issues associated with renewable energy output [17,18]. Meanwhile, energy storage proves effective in addressing the instability and intermittency of energy supply [19]. With the advancement of diverse energy storage technologies, energy storage transitions from a singular type to the coordination of multiple types. New types of energy storage [20], represented by HSS, are gaining widespread attention.

Therefore, by constructing a multi-energy complementarity system along with multi-energy storage, the instability caused by source-load uncertainty in system operation can be effectively mitigated.

Numerous studies have been conducted on the multi-time scale scheduling problems of multi-energy complementarity system along with energy storage. Aiming to maximize the benefit and profit of a wind-water-solar-pumped-storage-hybrid power system, a two-stage optimization scheduling model is proposed in [21]. In addition, a complementary wind-solar-hydro-storage system is established in [22], introducing a three-stage model for the daily generation plan. The proposed methods in [21,22] improve DERs utilization efficiency and reduces curtailment rates. Moreover, within a DS characterized by a substantial share of renewable energy sources, hydrogen-based energy systems exhibit the capability to absorb surplus renewable energy from sources such as photovoltaic, wind and hydro, releasing it when demand arises. This inherent flexibility facilitates enhanced integration and utilization of hydrogen-based energy systems within the DS [23]. A novel electro-hydrogen energy system model is established in [24] and it conducts an economic analysis of coupled energy storage systems. The introduction of a hydrogen-based power generation system for long-term stable operation can effectively enhance the flexibility and reliability of MG. Utilizing power-to-hydrogen (P2H) technology to convert surplus electricity into hydrogen for backup purposes, and employing hydrogen-to-power (H2P) technology during high loads to convert hydrogen back into electricity, mitigates wind and solar curtailment rates, thereby increasing overall energy utilization. [25] indicates that electrical energy storage (EES) [26] is more suitable for short-term electricity storage, with limitations on scale and reliability for prolonged operation, highlighting the cost advantages of hydrogen energy. Therefore, combining hydrogen energy with wind-solar-hydro-storage system will be advantageous in addressing electricity quantity balance issues across multiple time scales.

As HSS functions as a long-duration storage device, in order to fully leverage the advantages of hydrogen energy in the DS, it is essential to coordinate HSS at multi-time scale. However, existing research on multi-timescale studies mostly focuses on solving scheduling strategies within short time scales, such as day-ahead to intra-day, with limited studies analyzing the coordination between DERs with different timescale regulation characteristics, particularly from the perspective of the regulatory dynamics of DERs operating at various time scales.

(2) Multi spatial coordinated scheduling: The multi spatial coordinated scheduling problem encounters challenges both in decision-making privacy and computational tractability [27]. As the number of flexible resources in DS increases, the scheduling for MMG system becomes difficult to solve in a centralized approach. With the rise in the number of control variables, centralized optimization may face the "curse of dimensionality". Moreover, as the distribution system operator (DSO) and MGs normally have differentiated optimization objectives, it is also infeasible to formulate a centralized scheduling strategy from a global perspective. Consequently, the coordinated scheduling of MMG system is transitioning from centralized to distributed algorithms, and the decision-making mechanism is shifting from a single mode to autonomous and coordinated modes. Numerous studies have been conducted on the distributed scheduling problems of MMG systems. A fully alternating direction method of multipliers (ADMM) based distributed reactive power optimization algorithm is proposed in [28] to obtain the global optimum solution, effectively protecting the privacy of each entity. [29] employs stochastic probability modeling for small-scale energy resources and load demands in each MG based on a coordinated scheduling framework for MMG systems. The objective is to minimize the operating cost of the MMG system, and a particle swarm algorithm is utilized to solve for the operating strategies of each MG. Moreover, a distributed multi-energy management framework for the coordinated operation of interconnected biogas-solar-wind MMG systems is proposed in [30]. And based on a data-driven distributed robust model predictive control approach, a multi-time scale hydrogen-based MMG energy management framework capable of effectively handling the uncertainties in renewable energy and load is presented in [31]. In addition, [32] interconnects MGs with different energy characteristics and establishes a MMG coordinated scheduling model based on the federated demand response plan using forecasted average voting.

The aforementioned scheduling models for MMG systems provide economically viable solutions for optimization and scheduling. However, the aforementioned models do not consider the benefits allocation between MGs and DS. MG, as an independent entity, should share the benefits according to its contribution to the power and energy balance. In addition, existing methods on scheduling strategies for MMG systems often overlook the impact of MG's topology. As a large number of DERs connect to MGs, significant changes in MG power flow distribution and system losses may occur. Neglecting the consideration of the network topology may lead to issues such as node voltage violations, current and power exceedances, or excessive network losses.

In summary, in existing research on multi temporal scheduling, few studies consider the impact of long-duration energy storage. Additionally, multi spatial scheduling must address privacy concerns and balance the interests of different entities. Therefore, there still is a research gap in the field of coordinated scheduling for MMG systems that simultaneously considers both temporal and spatial perspectives, incorporating long-duration energy storage.

1.3. Contributions and paper organization

The scientific aim of the work is to explore the regulatory potential of MMG systems with both short-duration and long-duration energy storage in addressing the local consumption issue of high-proportion renewable energy across multi-time scales. The subject of the research focus on the multi-time coordinated scheduling problem of multi-energy complementary MMG system, aiming to address the operational challenges posed by the large-scale integration of DERs into DS. This study extends the existing research in the field by proposing a novel three-stage energy management framework for MMG systems considering the uncertainty renewable energy and long-duration energy storage. The major contributions of this paper can be summarized as follows:

1. A comprehensive multi-energy complementary energy management framework and a multi-time scale rolling optimization scheduling strategy for MMG systems considering long-duration energy storage, are proposed from both temporal and spatial perspectives, which enhances resource utilization efficiency and improves local absorption of renewable energy.

2. The rolling optimization and chance constraints are employed to mitigate the impact of source-load uncertainty at different time scales. And a three-stage electricity power and energy balance model incorporating long-duration energy storage is proposed, which fills the gap in the study of coordinated scheduling strategies for MMG system with long-duration energy storage across time scales from month to day.

3. The MG topology is considered in the model and the benefits allocation between MGs and DS is achieved based on their respective contributions to the power and energy balance. And considering the privacy protection, a variable step-size ADMM algorithm is employed to achieve distributed solutions.

Briefly, this article brings a new look to the existing literature in the following areas: (I) Innovative multi-time-scale scheduling strategy for coordinating short-duration and long-duration energy storage with multiple energy forms (II) Uncertainty optimization framework combining rolling optimization with chance constraints, (III) Benefits allocation mechanism considering the balance contribution of each MG, and (IV) Distributed solution technology integrated with a variable step-size ADMM algorithm.

The remaining sections of the paper are organized as follows: Section 2 constructs the multi-time scale optimization scheduling strategy for DS with multi-microgrid; Section 3 introduces a three-stage model to deal with the multi-time scale scheduling problem, and proposes a cost allocation method using Shapley value method for DS with MMG; Section 4 introduces the model solution method; Sections 5 and 6 give the case study and conclusions.

2. Multi-time scale optimization scheduling strategy for distribution system with multi-microgrid

An optimization scheduling strategy is proposed in this section. The scheduling strategy presents a coordinated management approach for multiple energy sources including wind, solar, water, and hydrogen. Furthermore, the scheduling strategy achieves electric energy and power balance and cost allocation among DS with MMG through a distributed management architecture from a spatial perspective. From a temporal perspective, it employs rolling optimization to coordinate scheduling across multi-time scales, including month, week, and day, thereby enhancing the flexibility of resource utilization.

2.1. Multi-microgrid system coordinated scheduling strategy

The topological structure of multi-energy complementary MG is shown in Fig. 1. The power sources in the MG includes hydropower station (HS), photovoltaics (PV) and wind turbine generator (WTG). Due to the inherent variability and intermittency in the output of renewable energy sources, it is imperative to integrate energy storage systems to mitigate power fluctuations and enhance power supply quality. Given that the issue of power-quantity balance encompasses multiple time scales, this paper introduces a hybrid energy storage system incorporating electrochemical energy storage, hydropower system, and hydrogen system. This integrated system is designed to address the diverse regulatory requirements across various time scales, encompassing both electric power and energy considerations.

The power flow, hydrogen flow, and water flow are illustrated in Fig. 1. In the power flow, PVs, WTGs, HS, and H2P serve as power sources to supply electricity to the loads and P2H in the MG. EES can function both as a power source and as a load for regulation. In hydrogen flow, hydrogen is generated from the P2H device, and the produced hydrogen can be supplied to factories. When there is surplus electric energy, the hydrogen is supplied to the HSS, and when there is a shortage of electric power in the system, the H2P device can convert the hydrogen in the HSS into electric energy [33,34]. Similarly, in water flow, hydropower system is designed to adjust their electricity generation in response to an excess of renewable energy production by reducing output and storing water in reservoirs. Conversely, when there is insufficient renewable energy generation, the hydropower system increases its output by converting gravitational potential energy into electric energy. Hydrogen energy system and hydropower systems work in conjunction to ensure a longer-term balance between energy supply and demand, while EES is employed to guarantee short-term balance in power supply and demand within the system. Furthermore, if the MG experiences a supply–demand imbalance, it has the option to purchase electricity from the distribution grid to achieve power equilibrium. This not only enhances operational control and system stability but also realizes an operational mechanism of self-sufficiency, surplus power feed-in, and grid adjustment.

The multi-energy complementary MG is a complex infrastructure comprising various equipment and energy conversion processes. The MG also involves the coordinated management of energy storage units with different time-scale adjustment characteristics, especially for MMG system. Therefore, addressing the scheduling issues in multi-energy complementary MG should thoroughly consider the impact of network topology, ensuring that MGs maintain voltage, current, and power within limits during operation, while minimizing internal network losses. Operators of such systems must develop effective dispatching strategies to enhance both economic efficiency and reliability.

To achieve effective resource management, this paper adopts a two-tier management framework, which is shown in Fig. 2:

(1) Upper level: The upper level comprises the DS and its energy management system (EMS-DS). The EMS-DS is capable of acquiring operational scheduling data for large-scale resources within the area (primarily including HS, PV and WTG), as well as overall network load data. It is responsible for formulating centralized resource scheduling strategy for the control area and facilitating information exchange between MGs. EMS-DS collects operational data from lower-level EMS-MGs, establishes global optimization objectives, and communicates centralized resource scheduling strategy to EMS-MGs. Additionally, EMS-DS is responsible for gathering boundary information from MGs, consolidating and distributing order to each EMS-MG. This ensures power balance within the entire substation and facilitates decentralized coordination and interaction between MGs.

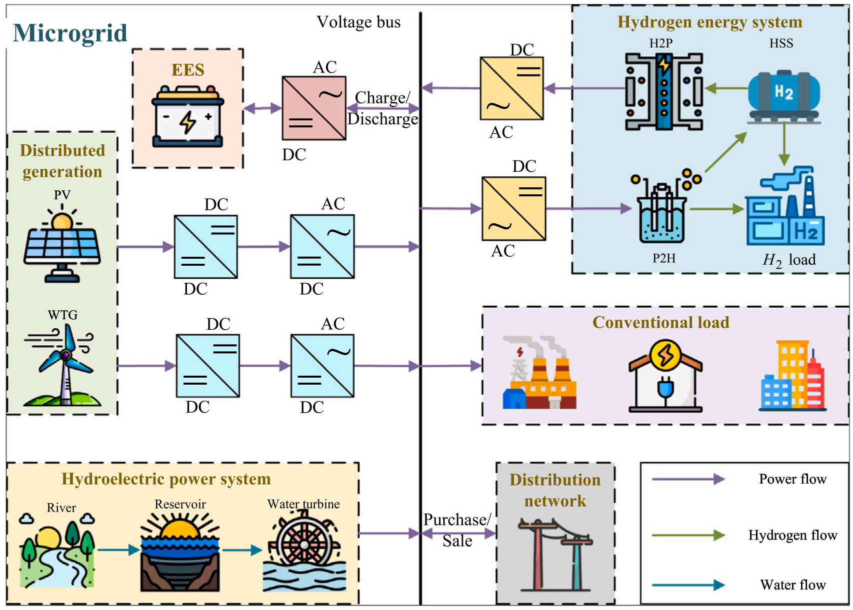


Fig. 1. Topological structure of wind-solar-hydro-hydrogen multi-energy complementary MG.

(2) Lower level: The lower level consists of MGs and their corresponding energy management systems (EMS-MG). The entire DS is coordinated and scheduled through a centralized EMS-DS and multiple distributed EMS-MGs. In addition, EMS-MG is responsible for collecting boundary information and scheduling orders transmitted by the EMS-DS, providing feedback information required by EMS-DS. EMS-MG formulates internal scheduling strategies and issues scheduling orders to the distributed unit devices, achieving power balance within MG and enabling power mutual assistance for other MGs, thereby realizing the optimization goals set by EMS-DS. Ultimately, the EMS-MG reports the scheduling costs to the EMS-DS. The EMS-DS then aggregates the total scheduling costs and allocates these costs using Shapley value method based on each MG's contribution to the power and energy balance.

2.2. Multi-time scale coordinated scheduling strategy

Due to the fluctuating nature of renewable energy output, an imbalance in electricity generation occurs across different time scales. Flexible resources such as HSS and HS, characterized by adjustable and substantial capacities with extended charge–discharge durations, are suitable for addressing long-term electricity balance issues within the system. On the other hand, EES, with its rapid response but shorter charge–discharge durations, is well-suited for participating in short-term electricity balance issues within the system. Therefore, the HSS is adopted a monthly balance period. Due to the regulatory capabilities of HS encompassing multiple time scales, such as daily and weekly intervals, this study introduces a scaling factor to enhance the flexibility of HS scheduling. The scaling factor is employed to partition the reservoir capacity of HS into weekly and daily components.

The introduction of long-duration energy storage systems (such as HS and HSS) necessitates the determination of their daily operational boundaries through the solution of long-period electric power and energy balance. However, due to the challenging task of providing accurate hourly-level output predictions owing to the precision limitations in renewable energy and load forecasting [35,36], and considering that the scheduling of large-capacity energy storage devices such as HS and HSS primarily addresses energy balance issues, this paper employs an electric energy balance model for analysis at the monthly and weekly time scales. At the daily time scale, an electric power balance model is utilized for further examination.

Based on the above analysis, the multi-time scale scheduling strategy framework in this study is illustrated in Fig. 3, encompassing three temporal scales: month, week, and day. The monthly electric energy balance (MEEB) provides the weekly operational boundaries for long-duration energy storage, while the weekly electric energy balance (WEEB), in turn, establishes the daily operational boundaries for long-duration energy storage in the daily electricity power balance (DEPB). The details are as follows:

(1) Month: At the month time scale, this study employs adjustable storage capacities of HSS and partially HS to achieve inter-week complementarity in electricity generation. In detailed, inputting the electric energy forecast data for the upcoming month, the MEEB model is solved to obtain the initial and final capacity states of HSS and HS (the portion utilized for MEEB) for each week. The weekly operational boundaries is then transmitted to the WEEB model.

(2) Week: At the week time scale, the remaining adjustable storage capacity of HS is utilized for inter-day complementarity. In detailed, the input consists of electric energy forecast data for the upcoming week. By solving the WEEB model, the initial and final capacity states of HSS and HS for each day are obtained. The daily operational boundaries is then passed on to the DEPB model.

(3) Day: At day time scale, EES is employed to complement power within the day. In detailed, the input consists of electric power forecast data for the upcoming day. By solving the DEPB model, the day-ahead scheduling strategy of DERs of MMG system can be obtained.

The model of the proposed multi-time scale scheduling strategy framework can be described as:

obj:

$$\min F \tag{1}$$

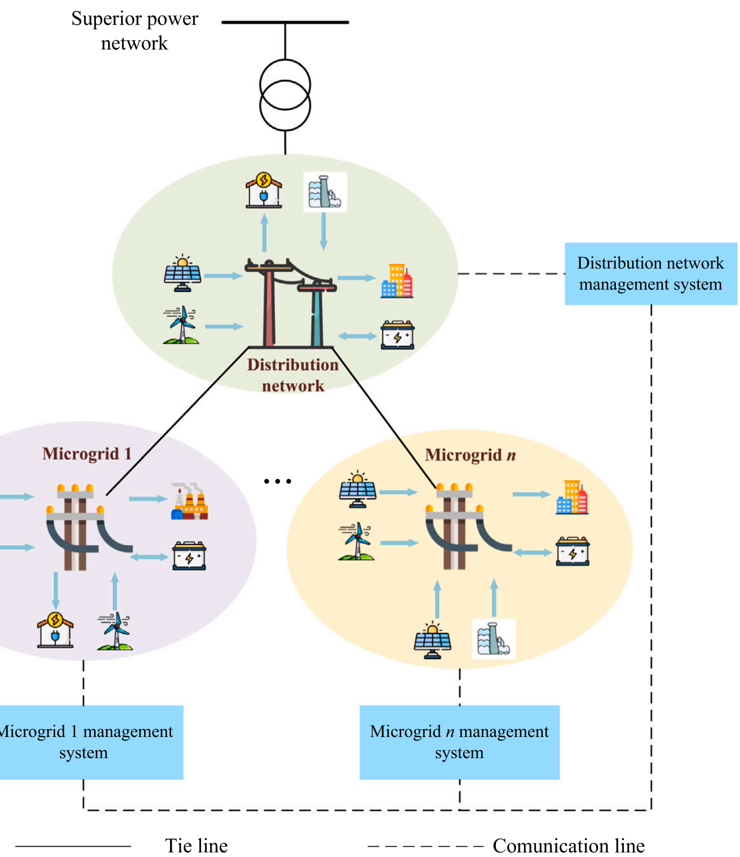


Fig. 2. Topological structure of wind-solar-hydro-hydrogen multi-energy complementary MG.

s.t.:

$$\boldsymbol{\beta} \boldsymbol{X}_{\mathrm{DER}} \leq \boldsymbol{\gamma} \tag{2}$$

$$\sum_{i \in \Omega_{\mathrm{DER}}, \tau \in \Gamma} \boldsymbol{X}_{\mathrm{DER}, i, \tau} = \mathbf{X}_{\mathrm{LOAD}, \tau} \tag{3}$$

$$\mathbf{SOC}_{j,0} = \mathbf{SOC}_{j,\mathrm{initial}}, \ j \in \Omega_{\mathrm{B}} \tag{4}$$

$$\mathbf{SOC}_{j,1} = \mathbf{SOC}_{j,\mathrm{final}}, \ j \in \Omega_{\mathrm{B}} \tag{5}$$

where F is the objective function which includes operating costs and curtailment penalty costs, with the DEPB model also incorporating network loss penalty costs. Eq. (2) represents the output constraints of DERs. $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ represent the coefficient matrix of the output of DERs $\boldsymbol{X}_{\mathrm{DER}}$. Eq. (3) represents the power and energy balance constraints. $\boldsymbol{X}_{\mathrm{DER},\tau}$ and $\boldsymbol{X}_{\mathrm{LOAD},\tau}$ represent the output of DERs and load at τ, respectively, while τ denotes the time scale of the model. For the MEEB, WEEB, and DEPB models, τ corresponds to $\boldsymbol{W}$, $\boldsymbol{D}$, and $\boldsymbol{T}$, respectively. $\boldsymbol{W}$, $\boldsymbol{D}$, and $\boldsymbol{T}$ are the collection of weeks in the month, days in the week and hours in the day, respectively. Eqs. (4) and (5) represent the initial and final energy state constraints of storage resources. $\mathbf{SOC}_{j,0}$ and $\mathbf{SOC}_{j,1}$ represent the initial and final energy states of storage devices, respectively. Ω_{DER} and Ω_{B} denote the sets of DER and storage devices, respectively, while $\mathbf{SOC}_{j,\mathrm{initial}}$ and $\mathbf{SOC}_{j,\mathrm{final}}$ represent the reference values for the initial and final energy states.

It is worth noting that in the aforementioned process, this study incorporates the concept of rolling optimization to mitigate the impact of uncertainties arising from electric energy forecasting. As shown in Fig. 3, the specific approach involves the scrolling execution of the MEEB model on a weekly basis to iteratively adjust the energy storage capacity's initial and final states. The WEEB model solely utilizes the monthly electric energy balance results obtained from the current week, while the daily electric power balance model is executed on a daily basis, relying on the weekly electric energy balance results from the current day for its computations. In addition, we have incorporated considerations of the uncertainty of renewable energy into the process of scroll optimization and the fuzzy chance constraints is employed to describe uncertainty. The accuracy of renewable energy forecasts generally increases with a shorter time scale [37]. As the time scale shortens, the reliability requirements of the system increase, necessitating a higher confidence level to ensure the stable operation of the system. Therefore, the confidence level α employed in this study also varies accordingly, as shown in Fig. 3, where $\alpha_1 > \alpha_2 > \alpha_3$.

EMS-DS can establish electric energy balance models, at month and week time scales based on the data reported by EMS-MG. Therefore, these models can be solved using a centralized control mode in EMS-DS. Since EES does not play a role in balancing electric energy at the daily and weekly scales, it can be ignored in these models. Moreover, according to the MMG framework established in Section 2.1, MGs and DS interact with each other through EMS-DS by exchanging only interconnection line power and current information. This enables power assistance and electric power and energy balance between MGs and DS.

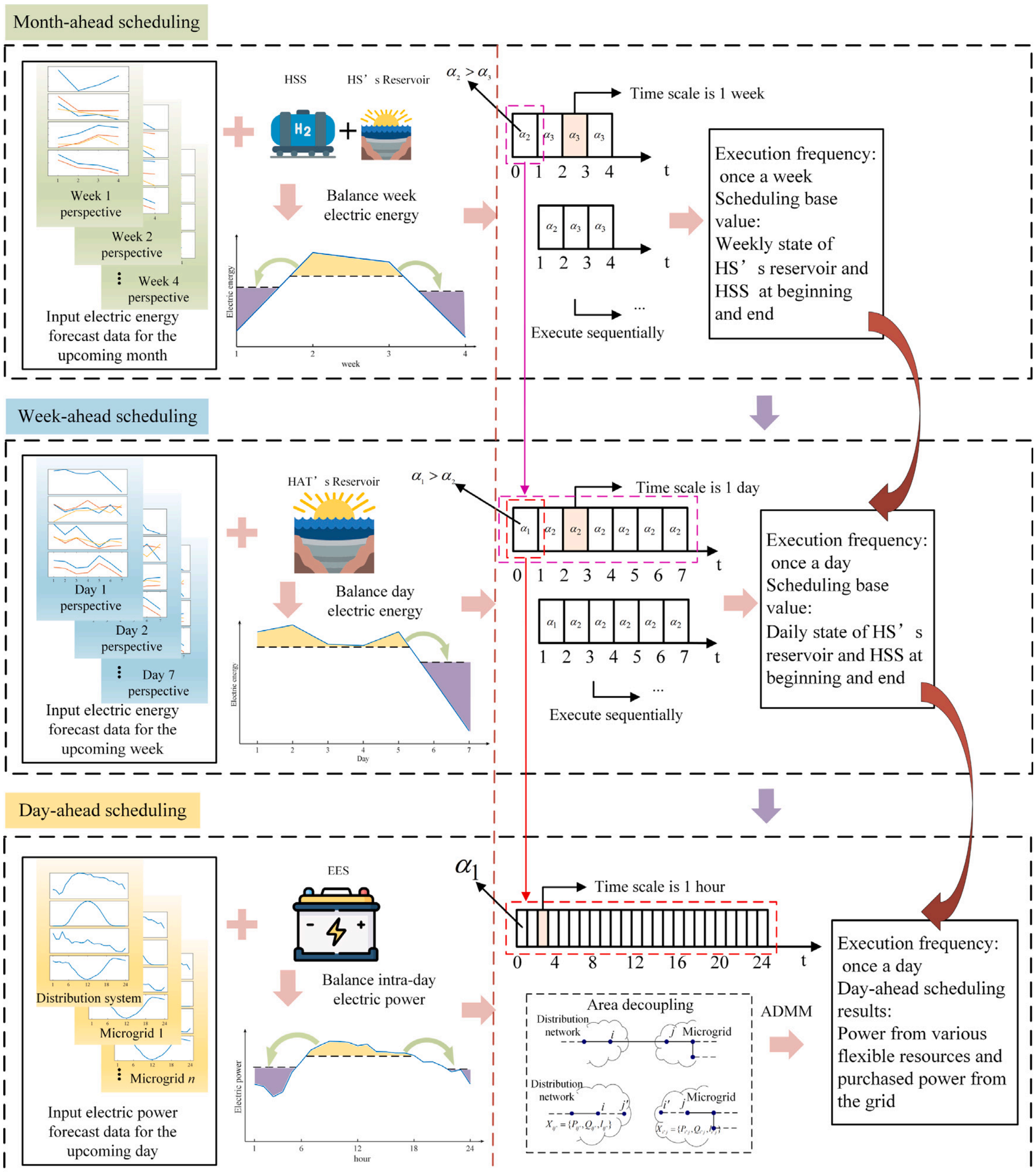


Fig. 3. Multi-time scale coordinated scheduling strategy framework.

3. Mathematical formulations

A three-stage electric energy and power balance model is proposed in this section. The interrelation among the three models is described in Section 2.2. The MEEB model and WEEB model incorporate equipment such as H2P, P2H, HSS, HS, WTG and PV, with constraints including energy upper and lower limits as well as time coupling constraints. These models are formulated as linear programming (LP) problems with fuzzy parameters. The DEPB model includes equipment such as H2P, P2H, HSS, HS, EES, WTG and PV, with constraints comprising power upper and lower limits, time coupling constraints, power flow constraints, and constraints involving 0–1 variables. This model is formulated as a mixed integer second-order cone program (MISOCP) with fuzzy parameters. The aforementioned models can all be solved by appropriate transformation and substitution into commercial solver.

3.1. Monthly electric energy balance model

In MEEB and WEEB model, EMS-DS receives power generation and load forecast data from EMS-MGs. It solves the output of the HSS and HS within the region, obtaining the initial and final hydrogen storage and water storage levels of these resources. Assuming that the collections of HSSs, HSs, WTGs, PVs, and EESs within the research region are denoted as $N_{\rm hss}$, $N_{\rm hs}$, $N_{\rm wtg}$, $N_{\rm pv}$ and $N_{\rm ees}$ respectively.

A. Objective Function

The objective function for MEEB model is formulated as the minimization of operational costs F_{c} and the maximization of renewable energy output F_{re}, and can be expressed as:

$$\min \sum_{w\in \boldsymbol{W}} F_{\mathrm{c}}(w) + \max \sum_{w\in \boldsymbol{W}} F_{\mathrm{re}}(w) \tag{6}$$

where F_{c} and F_{re} are as followed:

$$F_{\mathrm{c}}(w) = \sum_{n_{\mathrm{hs}}\in N_{\mathrm{hs}}} C_{\mathrm{hs},n_{\mathrm{hs}}} E_{\mathrm{hs},n_{\mathrm{hs}}}(w) + \sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}} [C_{\mathrm{h2p},n_{\mathrm{h}}} E_{\mathrm{h2p},n_{\mathrm{h}}}(w) + C_{\mathrm{p2h},n_{\mathrm{h}}} E_{\mathrm{h},n_{\mathrm{h}}}(w)] + C_{\mathrm{up}} E_{\mathrm{up}}(w) \tag{7}$$

$$F_{\mathrm{re}}(w) = \sum_{n_{\mathrm{hs}}\in N_{\mathrm{hs}}} \rho_{\mathrm{cle,hs}} E_{\mathrm{hs},n_{\mathrm{hs}}}(w) + \sum_{n_{\mathrm{pv}}\in N_{\mathrm{pv}}} \rho_{\mathrm{cle,pv}} E_{\mathrm{pv},n_{\mathrm{pv}}}(w) + \sum_{n_{\mathrm{wtg}}\in N_{\mathrm{wtg}}} \rho_{\mathrm{cle,wtg}} E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w) \tag{8}$$

where C_{x,n_x} and E_{x,n_x} represent the cost coefficient of energy and electric energy of n_xth flexible resource x, $\rho_{\mathrm{cle,hs}}$, $\rho_{\mathrm{cle,pv}}$ and $\rho_{\mathrm{cle,wtg}}$ represent the curtailment cost of electric energy for HS, PV, WTG, respectively. C_{up} represents the unit purchasing cost from the higher-level power grid and E_{up} represents the transmitted power from the higher-level power grid.

B. Constraints

(1) Hydrogen system:

The power from electricity devices can be converted to hydrogen through electrolysis (P2H), and hydrogen can be converted to power through fuel cells (H2P). These processes can be described as [38]:

$$H_{\mathrm{p2h},n_{\mathrm{h}}}(w) = \frac{\eta_{\mathrm{p2h}} CF_{\mathrm{H}}}{HHV_{\mathrm{H}}} P_{\mathrm{p2h},n_{\mathrm{h}}}(w) \tag{9}$$

$$P_{\mathrm{h2p},n_{\mathrm{h}}}(w) = \eta_{\mathrm{h2p}} LHV_{\mathrm{H}} H_{\mathrm{h2p},n_{\mathrm{h}}}(w) \tag{10}$$

where $H_{\mathrm{p2h},n_{\mathrm{h}}}$ is the hydrogen production for P2H device, and $P_{\mathrm{h2p},n_{\mathrm{h}}}$ is the power generation for H2P device. η_{p2h} and η_{h2p} respectively represent the P2H and H2P unit efficiency. HHV_{H} and LHV_{H} represent higher and lower heating value of hydrogen. CF_{H} is hydrogen compressibility factor. The hydrogen produced by the P2H device is partially allocated for supplying the hydrogen load $H_{\mathrm{load},n_{\mathrm{h}}}$, while a portion thereof can be stored in HSS. The storage capacity status of HSS $SOC_{\mathrm{hss},n_{\mathrm{h}}}$ can be characterized as follows:

$$SOC_{\mathrm{hss},n_{\mathrm{h}}}(t) = SOC_{\mathrm{hss},n_{\mathrm{h}}}(w-1) + H_{\mathrm{p2h},n_{\mathrm{h}}}(w) - H_{\mathrm{h2p},n_{\mathrm{h}}}(w) - H_{\mathrm{load},n_{\mathrm{h}}}(w) \tag{11}$$

Therefore, the constraints of hydrogen system can be described as:

$$0 \le E_{\mathrm{p2h},n_{\mathrm{h}}}(w) \le \sum_{w\in W} P_{\mathrm{p2h},n_{\mathrm{h}},\max} \Delta w \tag{12}$$

$$0 \le E_{\mathrm{h2p},n_{\mathrm{h}}}(w) \le \sum_{w\in W} P_{\mathrm{h2p},n_{\mathrm{h}},\max} \Delta w \tag{13}$$

$$SOC_{\mathrm{hss},n_{\mathrm{h}}}(w+\Delta w) = SOC_{\mathrm{hss},n_{\mathrm{h}}}(w) + \left(\frac{\eta_{\mathrm{p2h}} CF_{\mathrm{H}}}{HHV_{\mathrm{H}}} E_{\mathrm{p2h},n_{\mathrm{h}}}(w) - \frac{E_{\mathrm{h2p},n_{\mathrm{h}}}(w)}{\eta_{\mathrm{h2p}} LHV_{\mathrm{H}}} \right) \Delta w - H_{\mathrm{load},n_{\mathrm{h}}}(w) \tag{14}$$

$$SOC_{\mathrm{hss},n_{\mathrm{h}},\min} \le SOC_{\mathrm{hss},n_{\mathrm{h}}}(w) \le SOC_{\mathrm{hss},n_{\mathrm{h}},\max} \tag{15}$$

$$SOC_{\mathrm{hss},n_{\mathrm{h}}}(0) = SOC_{\mathrm{hss},n_{\mathrm{h}}}(4) \tag{16}$$

where $E_{\mathrm{p2h},n_{\mathrm{h}}}$ and $E_{\mathrm{h2p},n_{\mathrm{h}}}$ respectively represent electric energy consumption and generation of hydrogen system. And $SOC_{\mathrm{hss},n_{\mathrm{h}},\min}$ and $SOC_{\mathrm{hss},n_{\mathrm{h}},\max}$ respectively represent the minimum and maximum $SOC_{\mathrm{hss},n_{\mathrm{h}}}$.

(2) HS:

The output of a HS can be expressed as:

$$P_{\mathrm{hs},n_{\mathrm{hs}}} = g \eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}} Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}} \tag{17}$$

where g is acceleration of gravity, $\eta_{\mathrm{hs},n_{\mathrm{hs}}}$ is the efficiency of hydraulic turbine, $H_{\mathrm{hs},n_{\mathrm{hs}}}$ is the height of water head of HS, $Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}$ is forecast inflow. The constraints of HS can be described as:

$$0 \le E_{\mathrm{hs},n_{\mathrm{hs}}}(w) \le \sum_{w\in W} P_{\mathrm{hs},n_{\mathrm{hs}},\max} \Delta w \tag{18}$$

$$V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(w+\Delta w) = V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(w) + Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w) - \frac{E_{\mathrm{hs},n_{\mathrm{hs}}}(w)}{g \eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}}} - V_{\mathrm{ab},n_{\mathrm{hs}}}(w) \tag{19}$$

$$0 \le V_{\mathrm{ab},n_{\mathrm{hs}}}(w) \le Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w) \tag{20}$$

$$V_{\mathrm{hs},n_{\mathrm{hs}},\min} \le V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(w) \le V^{m}_{\mathrm{hs},n_{\mathrm{hs}},\max} \tag{21}$$

$$V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(0) = V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(4) = V^{m,0}_{\mathrm{hs},n_{\mathrm{hs}}} \tag{22}$$

where $P_{\mathrm{hs},n_{\mathrm{hs}},\max}$ represents the maximum power generation of a HS station, $V^{m,0}_{\mathrm{hs},n_{\mathrm{hs}}}$ is the monthly initial storage value. $V^{w}_{\mathrm{hs},n_{\mathrm{hs}}}$ denotes the reservoir storage capacity in month time scale, and the total reservoir storage capacity is the sum of monthly and weekly reservoir storage capacity:

$$V_{\mathrm{hs},n_{\mathrm{hs}}} = V^{m}_{\mathrm{hs},n_{\mathrm{hs}}} + V^{w}_{\mathrm{hs},n_{\mathrm{hs}}} \tag{23}$$

$V^m_{\mathrm{hs},n_{\mathrm{hs}},\max}$ represents the maximum $V^m_{\mathrm{hs},n_{\mathrm{hs}}}$, which can be obtained as:

$$V^m_{\mathrm{hs},n_{\mathrm{hs}},\max}=\sigma^m_{\mathrm{hs},n_{\mathrm{hs}}}V_{\mathrm{hs},n_{\mathrm{hs}},\max} \tag{24}$$

where $V_{\mathrm{hs},n_{\mathrm{hs}},\min}$ and $V_{\mathrm{hs},n_{\mathrm{hs}},\max}$ represent the minimum and maximum reservoir storage capacity for the HS station. $\sigma^m_{\mathrm{hs},n_{\mathrm{hs}}}$ is scaling factor, which can be manually set according to the scheduling plan. $V_{\mathrm{ab},n_{\mathrm{hs}}}(w)$ denotes the spillage of water for the HS station over the duration of week w.

(3) PV and WTG

The output of PV and WTG can be adjusted below the forecasted data, which can be described as:

$$0\le E_{\mathrm{pv},n_{\mathrm{pv}}}(w)\le E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(w) \tag{25}$$

$$0\le E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)\le E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(w) \tag{26}$$

where the superscript 'fore' denotes forecasted data. As the focus of this study does not lie in source-load forecasting methodologies, the forecasted values in this paper are assumed to be known quantities.

(4) Transmitted power to the higher-level power grid

$$0\le E_{\mathrm{up}}(w)\le \sum_{w\in \mathrm{W}} P_{\mathrm{up,max}}\Delta w \tag{27}$$

where $P_{\mathrm{up.\,max}}$ represents the maximum power transferable from the higher-level power grid.

(5) Electric energy balance and reserve

The electric energy balance constraint is:

$$\begin{aligned}&E_{\mathrm{up}}(w)+\sum_{n_h\in N_h}\left[E_{\mathrm{h2p},n_{\mathrm{h}}}(w)-E_{\mathrm{p2h},n_{\mathrm{h}}}(w)-E_{h,n_h}(w)\right]+\sum_{n_{pv}\in N_{pv}}E_{\mathrm{pv},n_{\mathrm{pv}}}(w)\\&\quad+\sum_{n_{wtg}\in N_{wtg}}E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)+\sum_{n_{hs}\in N_{hs}}E_{\mathrm{hs},n_{\mathrm{hs}}}(w)=E_{\mathrm{load}}(w)+\Delta E_{\mathrm{loss,week}}(w)\end{aligned} \tag{28}$$

where $E_{\mathrm{h},n_{\mathrm{h}}}$ and E_{load} represent the electric energy of hydrogen load and electricity load, and $\Delta E_{\mathrm{loss,week}}$ represents the average electric energy losses in week, which includes electric energy losses during the energy storage charging and discharging and network losses. $\Delta E_{\mathrm{loss,week}}$ can be obtained as:

$$\Delta E_{\mathrm{loss,week}}=\sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}}\left(\frac{HHV_{\mathrm{h}}}{\eta_{\mathrm{p2h}}CF_{\mathrm{h}}}-\eta_{\mathrm{h2p}}LHV_{\mathrm{h}}\right)\bar{H}_{\mathrm{h2p},n_{\mathrm{h}}}\cdot 7+\sum_{n_{\mathrm{ees}}\in N_{\mathrm{ees}}}(1-\eta^c_{n_{\mathrm{ees}}}\eta^d_{n_{\mathrm{ees}}})Q_{\mathrm{ees},n_{\mathrm{ees}}}f_{\mathrm{ees},n_{\mathrm{ees}}}\cdot 7+\bar{E}_{\mathrm{loss}}\cdot 7 \tag{29}$$

where $\bar{H}_{\mathrm{h2p},n_{\mathrm{h}}}$ is average hydrogen consumption per day of H2P device, $\eta^c_{n_{\mathrm{ees}}}$ and $\eta^d_{n_{\mathrm{ees}}}$ are the charge and discharge efficiency of EES device, $Q_{\mathrm{ees},n_{\mathrm{ees}}}$ is energy storage capacity and $f_{\mathrm{ees},n_{\mathrm{ees}}}$ is average daily charge and discharge frequency. $\bar{E}_{\mathrm{loss}}$ is average network loss. In the operation of the power grid, a certain amount of reserve capacity needs to be established. In this paper, the selection for the reserve power source is the higher-level power grid and HS. Thus, the electric energy reserve can be described as:

$$[\sum_{d\in D}P_{\mathrm{up,max}}\Delta d-E_{\mathrm{up}}(w)]+\min[g\eta_{\mathrm{hs},n_{\mathrm{hs}}}H_{\mathrm{hs},n_{\mathrm{hs}}}V_{\mathrm{hs},n_{\mathrm{hs}}}(w+1),\sum_{d\in D}P_{\mathrm{hs},n_{\mathrm{hs}},\max}\Delta d]\ge \rho_{\mathrm{load},1}E_{\mathrm{load}}(w) \tag{30}$$

where $\rho_{\mathrm{load},1}$ represents the electric energy reserve coefficient in MEEB model.

(6) Fuzzy chance constraints

The uncertainty in energy sources and load forecasts is existed in aforementioned model. Both robust optimization and chance constraints programming address source-load uncertainty by increasing the conservatism of decision-making. However, the results obtained from robust optimization are typically conservative, and the derived scheduling plans may not be applicable in the actual operation of DS and MGs. By contrast, chance constraints programming ensures that stochastic constraints are satisfied at a given confidence level, effectively balancing the demands for both economic efficiency and robustness [39]. Therefore, the forecast source-load variable v^{fore} contained fuzzy variable χ can be expressed as:

$$v^{\mathrm{fore}}=\mathrm{E}(v^{\mathrm{fore}})+\chi \tag{31}$$

where $\mathrm{E}(\cdot)$ represents the expected value function. v^{fore} is the forecast source-load variable, which in this section is $Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w)$, $E_{\mathrm{pv},n_{\mathrm{pv}}}(w)$, $E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)$, $E_{\mathrm{load}}(w)$. The fuzzy theory is introduced to represent the forecast error of sources and load with fuzzy variables χ. The fuzzy variables of forecast error are denoted as $\Delta\tilde{Q}^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w)$, $\Delta\tilde{E}^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(w)$, $\Delta\tilde{E}^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)$ and $\Delta\tilde{E}_{\mathrm{load}}(w)$, respectively. The fuzzy variables lead to the inability to determine a specific feasible set for the constraint conditions. For the model involving uncertain fuzzy variables χ, the expression of the constraint is as follows:

$$Cr\{g(x,\chi)\le 0\}\ge\alpha \tag{32}$$

where $Cr\{\cdot\}$ means the credibility of the case in $\{\cdot\}$ and α is the confidence level. $g(x,\chi)$ is the constraint function. Therefore, Eq. (31) represents the constraints that hold with a certain confidence level α. The value of α is related to the time dimension of the electric energy balance model, and the setting principle can be seen in Section 2.2. x is the decision variable, which in this section is the amount of month variation $[V_{\mathrm{ab},n_{\mathrm{hs}}}(w), E_{\mathrm{pv},n_{\mathrm{pv}}}(w), E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w), V^m_{\mathrm{hs},n_{\mathrm{hs}}}(w), E_{\mathrm{hs},n_{\mathrm{hs}}}(w), E_{\mathrm{up}(w)}, E_{\mathrm{h2p},n_{\mathrm{h}}}(w), E_{\mathrm{p2h},n_{\mathrm{h}}}(w), E_{\mathrm{h},n_{\mathrm{h}}}(w)]$. Therefore, the chance constraints under credibility measures are employed for handling this uncertainty. Eqs. (19)–(20), (25)–(26), (28), (30) can be expressed in the form of fuzzy chance constraints:

$$Cr\{\mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w))+\Delta\tilde{Q}^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w)-V_{\mathrm{ab},n_{\mathrm{hs}}}(w)\ge 0\}\ge\alpha \tag{33}$$

$$Cr\{\mathrm{E}(E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(w))+\Delta\tilde{E}^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(w)-E_{\mathrm{pv},n_{\mathrm{pv}}}(w)\ge 0\}\ge\alpha \tag{34}$$

$$Cr\{\mathrm{E}(E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(w))+\Delta\tilde{E}^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)-E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)\ge 0\}\ge\alpha \tag{35}$$

$$Cr\left\{V^m_{\mathrm{hs},n_{\mathrm{hs}}}(w+\Delta w)=V^m_{\mathrm{hs},n_{\mathrm{hs}}}(w)+\mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w))+\Delta\tilde{Q}^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w)-\frac{E_{\mathrm{hs},n_{\mathrm{hs}}}(w)}{g\eta_{\mathrm{hs},n_{\mathrm{hs}}}H_{\mathrm{hs},n_{\mathrm{hs}}}}-V_{\mathrm{ab},n_{\mathrm{hs}}}(w)\right\}\geq\alpha \tag{36}$$

$$\begin{aligned}Cr\Bigg\{&E_{\mathrm{up}}(w)+\sum_{n_{\mathrm{pv}}\in N_{\mathrm{pv}}}E_{\mathrm{pv},n_{\mathrm{pv}}}(w)+\sum_{n_{\mathrm{wtg}}\in N_{\mathrm{wtg}}}E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)+\sum_{n_{\mathrm{hs}}\in N_{\mathrm{hs}}}E_{\mathrm{hs},n_{\mathrm{hs}}}(w)+\sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}}\Big[E_{\mathrm{h2p},n_{\mathrm{h}}}(w)-\\&E_{\mathrm{p2h},n_{\mathrm{h}}}(w)-E_{\mathrm{h},n_{\mathrm{h}}}(w)\Big]=\mathrm{E}(E_{\mathrm{load}}(w))+\Delta\tilde{E}_{\mathrm{load}}(w)+\Delta E_{\mathrm{loss,week}}(w)\Bigg\}\geq\alpha\end{aligned} \tag{37}$$

$$\begin{aligned}Cr\Bigg\{&\left[\sum_{d\in \boldsymbol{D}}P_{\mathrm{up,max}}\Delta d-E_{\mathrm{up}}(w)\right]+\min\left[g\eta_{\mathrm{hs},n_{\mathrm{hs}}}H_{\mathrm{hs},n_{\mathrm{hs}}}V_{\mathrm{hs},n_{\mathrm{hs}}}(w+1),\sum_{d\in \boldsymbol{D}}P_{\mathrm{hs},n_{\mathrm{hs}},\mathrm{max}}\Delta d\right]\\&\geq\rho_{\mathrm{load},1}\left(\mathrm{E}(E_{\mathrm{load}}(w))+\Delta\tilde{E}_{\mathrm{load}}(w)\right)\Bigg\}\geq\alpha\end{aligned} \tag{38}$$

3.2. Weekly electric energy balance model

In WEEB model, the electric energy quantity corresponds to daily energy. Partial boundary conditions for HS and HSS devices in the WEEB model are determined by the results of solving MEEB model. The WEEB model is fundamentally consistent with the MEEB model.

A. Objective function

$$\min\sum_{d\in \boldsymbol{D}}F_{\mathrm{c}}(d)+\max\sum_{d\in \boldsymbol{D}}F_{\mathrm{re}}(d) \tag{39}$$

B. Constraints

(1) P2H and H2P:

The constraints of hydrogen system can be described as Eqs. (12)–(15) (the time scale $\boldsymbol{W}$ changes to $\boldsymbol{D}$) and the weekly initial and final state constraints for HSS:

$$SOC_{\mathrm{hss},n_{\mathrm{h}}}(0)=SOC^{m,results}_{\mathrm{hss},n_{\mathrm{h}}}(w) \tag{40}$$

$$SOC_{\mathrm{hss},n_{\mathrm{h}}}(7)=SOC^{m,\mathrm{results}}_{\mathrm{hss},n_{\mathrm{h}}}(w+1) \tag{41}$$

where $SOC^{m,\mathrm{results}}_{\mathrm{hss},n_{\mathrm{h}}}(w)$ is the results of MEEB model in wth week.

(2) HS:

Similarly, the constraints of $E_{\mathrm{hs},n_{\mathrm{hs}}}$ and $V_{\mathrm{ab},n_{\mathrm{hs}}}$ can be described as Eqs. (18) and (20) (the time scale $\boldsymbol{W}$ changes to $\boldsymbol{D}$). The temporal coupling constraint, upper and lower boundary, and the daily initial and final state constraint of $V_{\mathrm{hs},n_{\mathrm{hs}}}$ can be respectively descried as:

$$V_{\mathrm{hs},n_{\mathrm{hs}}}(d+1)=V_{\mathrm{hs},n_{\mathrm{hs}}}(d)+Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(d)-\frac{E_{\mathrm{hs},n_{\mathrm{hs}}}(d)}{g\eta_{\mathrm{hs},n_{\mathrm{hs}}}H_{\mathrm{hs},n_{\mathrm{hs}}}}-V_{\mathrm{ab},n_{\mathrm{hs}}}(d) \tag{42}$$

$$0\leq V_{\mathrm{hs},n_{\mathrm{hs}}}(d)\leq V_{\mathrm{hs},n_{\mathrm{hs}},\mathrm{max}} \tag{43}$$

$$V^w_{\mathrm{hs},n_{\mathrm{hs}}}(0)=V^w_{\mathrm{hs},n_{\mathrm{hs}}}(7)=V^{w,0}_{\mathrm{hs},n_{\mathrm{hs}}} \tag{44}$$

where $V^{w,0}_{\mathrm{hs},n_{\mathrm{hs}}}$ is the weekly initial storage. Additionally, the week initial and final state constraint of $V^m_{\mathrm{hs},n_{\mathrm{hs}}}$ are based on the results of WEEB model:

$$V_{\mathrm{hs},n_{\mathrm{hs}}}(0)=V^{m,\mathrm{results}}_{\mathrm{hs},n_{\mathrm{hs}}}(w) \tag{45}$$

$$V_{\mathrm{hs},n_{\mathrm{hs}}}(7)=V^{m,\mathrm{results}}_{\mathrm{hs},n_{\mathrm{hs}}}(w+1) \tag{46}$$

where $V^{m,\mathrm{results}}_{\mathrm{hs},n_{\mathrm{hs}}}(w)$ is the results of MEEB model in wth week.

(3) PV, WTG and transmitted power to the higher-level power grid

The constraints of PV, WTG and transmitted power to the higher-level power grid can be described as Eqs. (25)–(27) (the time scale $\boldsymbol{W}$ changes to $\boldsymbol{D}$).

(4) Electric energy balance and reserve

The constraint of electric energy balance of WEEB model is:

$$\begin{aligned}&E_{\mathrm{up}}(d)+\sum_{n_{\mathrm{pv}}\in N_{\mathrm{pv}}}E_{\mathrm{pv},n_{\mathrm{pv}}}(d)+\sum_{n_{\mathrm{wtg}}\in N_{\mathrm{wtg}}}E_{\mathrm{wtg},n_{\mathrm{wtg}}}(d)+\sum_{n_{\mathrm{hs}}\in N_{\mathrm{hs}}}E_{\mathrm{hs},n_{\mathrm{hs}}}(d)+\\&\sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}}\Big[E_{\mathrm{h2p},n_{\mathrm{h}}}(d)-E_{\mathrm{p2h},n_{\mathrm{h}}}(d)-E_{\mathrm{h},n_{\mathrm{h}}}(d)\Big]=E_{\mathrm{load}}(d)+\Delta E_{\mathrm{loss,day}}(d)\end{aligned} \tag{47}$$

where $\Delta E_{\mathrm{loss,day}}$ represents the average electric energy losses in day, which can be obtained as:

$$\Delta E_{\mathrm{loss,day}}=\sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}}\left(\frac{HHV_{\mathrm{H}}}{\eta_{\mathrm{p2h}}CF_{\mathrm{H}}}-\eta_{\mathrm{h2p}}LHV_{\mathrm{H}}\right)\bar{H}_{\mathrm{h2p},n_{\mathrm{h}}}+\sum_{n_{\mathrm{ees}}\in N_{\mathrm{ees}}}(1-\eta^c_{n_{\mathrm{ees}}}\eta^d_{n_{\mathrm{ees}}})Q_{\mathrm{ees},n_{\mathrm{ees}}}f_{\mathrm{ees},n_{\mathrm{ees}}}+\bar{E}_{\mathrm{loss}} \tag{48}$$

The constraint of electric energy reserve of WEEB model is:

$$\left[\sum_{t\in \boldsymbol{T}}P_{up.\max}\Delta t-E_{up}(d)\right]+\min\left[g\eta_{\mathrm{hs},n_{\mathrm{hs}}}H_{\mathrm{hs},n_{\mathrm{hs}}}V_{\mathrm{hs},n_{\mathrm{hs}}}(d+1),\sum_{t\in \boldsymbol{T}}P_{\mathrm{hs},n_{\mathrm{hs}},\mathrm{max}}\Delta t\right]\geq\rho_{\mathrm{load},2}E_{\mathrm{load}}(d) \tag{49}$$

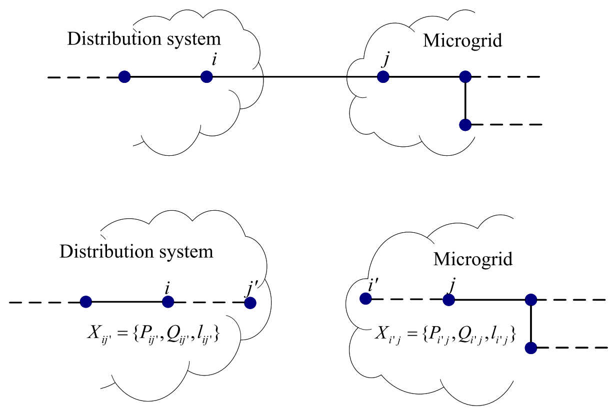


Fig. 4. Decoupling of tie lines.

where $\rho_{\text{load},2}$ represents the electric energy reserve coefficient in WEEB model.

(5) Fuzzy chance constraints

Similarly, v^{fore} in this section are $Q^{\text{fore}}_{\text{hs},n_{\text{hs}}}(d)$, $E_{\text{pv},n_{\text{pv}}}(d)$, $E_{\text{wtg},n_{\text{wtg}}}(d)$, $E_{\text{load}}(d)$. The fuzzy variables of forecast error are denoted as $\Delta\tilde{Q}^{\text{fore}}_{\text{hs},n_{\text{hs}}}(d)$, $\Delta\tilde{E}^{\text{fore}}_{\text{pv},n_{\text{pv}}}(d)$, $\Delta\tilde{E}^{\text{fore}}_{\text{wtg},n_{\text{wtg}}}(d)$ and $\Delta\tilde{E}_{\text{load}}(d)$, respectively. With reference to the MEEB model, the fuzzy chance constraints of the WEEB model can be formulated as:

$$Cr\{\mathrm{E}(Q^{\text{fore}}_{\text{hs},n_{\text{hs}}}(d)) + \Delta\tilde{Q}^{\text{fore}}_{\text{hs},n_{\text{hs}}}(d) - V_{\text{ab},n_{\text{hs}}}(d) \geq 0\} \geq \alpha \tag{50}$$

$$Cr\{\mathrm{E}(E^{\text{fore}}_{\text{pv},n_{\text{pv}}}(d)) + \Delta\tilde{E}^{\text{fore}}_{\text{pv},n_{\text{pv}}}(d) - E_{\text{pv},n_{\text{pv}}}(d) \geq 0\} \geq \alpha \tag{51}$$

$$Cr\{\mathrm{E}(E^{\text{fore}}_{\text{wtg},n_{\text{wtg}}}(d)) + \Delta\tilde{E}^{\text{fore}}_{\text{wtg},n_{\text{wtg}}}(d) - E_{\text{wtg},n_{\text{wtg}}}(d) \geq 0\} \geq \alpha \tag{52}$$

$$Cr\left\{ V_{\text{hs},n_{\text{hs}}}(d+1) = V_{\text{hs},n_{\text{hs}}}(d) + \mathrm{E}(Q^{\text{fore}}_{\text{hs},n_{\text{hs}}}(d)) + \Delta\tilde{Q}^{\text{fore}}_{\text{hs},n_{\text{hs}}}(d) - \frac{E_{\text{hs},n_{\text{hs}}}(d)}{g\eta_{\text{hs},n_{\text{hs}}} H_{\text{hs},n_{\text{hs}}}} - V_{\text{ab},n_{\text{hs}}}(d) \right\} \geq \alpha \tag{53}$$

$$\begin{aligned} Cr\Bigg\{ & E_{\text{up}}(d) + \sum_{n_{\text{pv}}\in N_{\text{pv}}} E_{\text{pv},n_{\text{pv}}}(d) + \sum_{n_{\text{wtg}}\in N_{\text{wtg}}} E_{\text{wtg},n_{\text{wtg}}}(d) + \sum_{n_{\text{hs}}\in N_{\text{hs}}} E_{\text{hs},n_{\text{hs}}}(d) + \\ & \sum_{n_{\text{h}}\in N_{\text{h}}} \left[E_{\text{h2p},n_{\text{h}}}(d) - E_{\text{p2h},n_{\text{h}}}(d) - E_{\text{h},n_{\text{h}}}(d) \right] = \mathrm{E}(E_{\text{load}}(d)) + \Delta\tilde{E}_{\text{load}}(d) + \Delta E_{\text{loss,day}}(d) \Bigg\} \geq \alpha \end{aligned} \tag{54}$$

$$\begin{aligned} Cr\Bigg\{ & \left[\sum_{t\in T} P_{\text{up, max}}\Delta t - E_{\text{up}}(d) \right] + \min\left[g\eta_{\text{hs},n_{\text{hs}}} H_{\text{hs},n_{\text{hs}}} V_{\text{hs},n_{\text{hs}}}(d+1), \sum_{t\in T} P_{\text{hs},n_{\text{hs}},\text{max}}\Delta t \right] \\ & \geq \rho_{\text{load},2} \left(\mathrm{E}(E_{\text{load}}(d)) + \Delta\tilde{E}_{\text{load}}(d) \right) \Bigg\} \geq \alpha \end{aligned} \tag{55}$$

It is worth mentioning that the confidence level α of the WEEB model should be higher than the confidence level α of the MEEB model. The setting principle also can be seen in Section 2.2.

3.3. Daily electric power balance model

The daily electric power balance (DEPB) model is conducted separately in EMS-MG and EMS-DS. In the realm of EMS-DS and EMS-MG, power equilibrium is achieved through the exchange of interconnectivity information. This ensures the preservation of user privacy within MGs. Thus, the first step involves decoupling the DS and the MGs. The decoupling diagrams of tie lines is illustrated in Fig. 4. As shown in the figure, the DS and MGs are interconnected through tie line ij in topology, where i and j represent boundary nodes of DS and MG, respectively. The first step involves disconnecting tie line ij and introducing replicated nodes i' and j' into the DS and the MG, respectively. These replicated nodes serve as load nodes and are connected to nodes j and i, forming virtual boundary tie lines ij' and $i'j$ for the DS and the MG. The decoupling is achieved by introducing the concept of replicated nodes, and the exchange of active power P_{ij}, reactive power Q_{ij}, and current l_{ij} (where l_{ij} represents the square of branch current) through tie lines is subjected to distributed computation between the DS and the MG. The voltage of replicated nodes can be calculated through power flow equations, thus establishing the consistency variable $X_{ij'} = \{P_{ij'}, Q_{ij'}, l_{ij'}\}$ for the DS relative to the MG, and $X_{i'j} = \{P_{i'j}, Q_{i'j}, l_{i'j}\}$ for the MG relative to the DS.

A. Objective function

For both DS and MG, the objective function is to minimize internal operational costs f_{c} and network losses f_{loss}, while maximizing renewable energy output f_{re}:

$$\min \sum_{t\in T} (f_{\text{c}}(t) + f_{\text{loss}}(t)) + \max \sum_{t\in T} f_{\text{re}}(t) \tag{56}$$

where f_c, f_{loss} and f_{re} can be expressed as:

$$\begin{cases} f_{\text{c}}(t) = f_{\text{up}}(t) + f_{\text{h}}(t) + f_{\text{hs}}(t) + f_{\text{ees}}(t) \\ f_{\text{loss}}(t) = \rho_{\text{loss}} P_{\text{loss}}(t) \\ f_{\text{re}}(t) = \sum\limits_{n_{\text{hs}} \in N_{\text{hs}}} \rho_{\text{cle,hs}} P_{\text{hs},n_{\text{hs}}}(t) + \sum\limits_{n_{\text{pv}} \in N_{\text{pv}}} \rho_{\text{cle,pv}} P_{\text{pv},n_{\text{pv}}}(t) + \sum\limits_{n_{\text{wtg}} \in N_{\text{wtg}}} \rho_{\text{cle,wtg}} P_{\text{wtg},n_{\text{wtg}}}(t) \end{cases} \tag{57}$$

where P_{loss} and ρ_{loss} represent the network losses power and corresponding cost coefficient, f_{up}, f_{h}, f_{hs} and f_{ees} represent the cost of corresponding variables, which can be expressed as:

$$\begin{cases} f_{\text{up}}(t) = c_{\text{up}}(t) P_{\text{up}}(t) \\ f_{\text{h}}(t) = \sum\limits_{n_{\text{h}} \in N_{\text{h}}} (c_{\text{h2p}} P_{\text{h2p},n_{\text{h}}}(t) + c_{\text{p2h}} P_{\text{p2h},n_{\text{h}}}(t)) \\ f_{\text{hs}}(t) = \sum\limits_{n_{\text{hs}} \in N_{\text{hs}}} c_{\text{hs},n_{\text{hs}}} P_{\text{hs},n_{\text{hs}}}(t) \\ f_{\text{ees}}(t) = \sum\limits_{n_{\text{ees}} \in N_{\text{ees}}} (c^c_{\text{ees},n_{\text{ees}}} P^c_{\text{ees},n_{\text{ees}}}(t) + c^d_{\text{ees},n_{\text{ees}}} P^d_{\text{ees},n_{\text{ees}}}(t)) \end{cases} \tag{58}$$

where c_{x,n_x} and P_{x,n_x} represent the cost coefficient of power and electric energy of n_xth flexible resource x, the superscript 'c' and 'd' in the variable of EES stands for charging and discharging, respectively. And the model of the MG does not include the variable f_{up}.

B. Constraints

In DEPB model, in addition to incorporating operational constraints on relevant variables as in MEEB model, distflow constraints considering network topology and operational constraints on EES have been introduced.

(1) Distflow

Distflow model [40] can be described as :

$$\begin{cases} V_j(t) = V_i(t) - 2(r_{ij} P_{ij}(t) + x_{ij} Q_{ij}(t)) + (r_{ij}^2 + x_{ij}^2) l_{ij}(t) \\ p_j(t) = P_{ij}(t) - r_{ij} l_{ij}(t) - \sum\limits_k P_{jk}(t) \\ q_j(t) = Q_{ij}(t) - x_{ij} l_{ij}(t) - \sum\limits_k Q_{jk}(t) \\ \left\| \begin{matrix} 2P_{ij}(t) \\ 2Q_{ij}(t) \\ V_i - l_{ij} \end{matrix} \right\|_2 \le V_i + l_{ij} \end{cases} \tag{59}$$

where $V_j(t) = U_j^2(t)$, $l_{ij}(t) = I_{ij}^2(t)$, U_j is the voltage at node j, I_{ij} is the current in branch ij, x_{ij} and r_{ij} are the reactance and resistance, respectively. k is the set of child nodes with node j as the parent node. The injection active power p_j and reactive power q_j of node j in distflow model can be expressed as:

$$\begin{cases} p_j(t) = P_{j,\text{load}}(t) + \sum\limits_{k_{\text{h}} \in \Omega_{\text{h},j}} (P_{\text{p2h},k_{\text{h}}}(t) - P_{\text{h2p},k_{\text{h}}}(t) - P_{\text{h},n_{\text{h}}}(t)) - \\ \sum\limits_{k_{\text{hs}} \in \Omega_{\text{hs},j}} P_{\text{hs},k_{\text{hs}}}(t) + \sum\limits_{k_{\text{ees}} \in \Omega_{\text{ees},j}} (P^c_{\text{ees},k_{\text{ees}}}(t) - P^d_{\text{ees},k_{\text{ees}}}(t)) - \\ \sum\limits_{k_{\text{pv}} \in \Omega_{\text{pv},j}} P_{\text{pv},k_{\text{pv}}}(t) - \sum\limits_{k_{\text{wtg}} \in \Omega_{\text{wtg},j}} P_{\text{wtg},k_{\text{wtg}}}(t) \\ q_j(t) = Q_{j,\text{load}}(t) - \sum\limits_{k_{\text{SVC}} \in \Omega_{\text{SVC},j}} Q_{\text{SVC},k_{\text{SVC}}}(t) \end{cases} \tag{60}$$

where $P_{j,\text{load}}$ and $Q_{j,\text{load}}$ represent the active and reactive load power at node j, respectively. $\Omega_{x,j}$ represents the collection of flexible resource x at node j. Additionally, there are constraints on line power, current, and node voltage:

$$\begin{cases} P_{ij} \le P_{ij,\max} \\ l_{ij} \le l_{ij,\max} \\ V_{i,\min} \le V_i \le V_{i,\max} \end{cases} \tag{61}$$

(2) HSS

The temporal coupling constraint of HSS is shown in Eq. (11). The upper and lower boundary of P2H, H2P and HSS can be expressed as:

$$0 \le P_{\text{p2h},n_{\text{h}}}(t) \le k_{n_{\text{hss}}} P_{\text{p2h},n_{\text{h}},\max} \tag{62}$$

$$0 \le P_{\text{h2p},n_{\text{h}}}(t) \le (1 - k_{n_{\text{hss}}}) P_{h2p,n_{\text{h}},\max} \tag{63}$$

$$SOC_{\text{hss},n_{\text{h}},\min} \le SOC_{\text{hss},n_{\text{h}}}(t) \le SOC_{\text{hss},n_{\text{h}},\max} \tag{64}$$

where $k_{n_{\text{hss}}}$ is a binary variable, 0 represents the H2P working, and 1 represents the P2H working. The daily initial and final state constraints for HSS:

$$SOC_{\text{hss},n_{\text{h}}}(0) = SOC^{w,\text{results}}_{\text{hss},n_{\text{h}}}(d) \tag{65}$$

$$SOC_{\text{hss},n_{\text{h}}}(24) = SOC_{\text{hss},n_{\text{h}}}^{w,\text{results}}(d+1) \tag{66}$$

where $SOC_{\text{hss},n_{\text{h}}}^{w,\text{results}}(d)$ is the results of MEEB model in dth day.

(3) HS

The constraints of $V_{\text{ab},n_{\text{hs}}}$ and $V_{\text{hs},n_{\text{hs}}}$ can be described as Eqs. (20) and (43) (time scale changes to $\boldsymbol{T}$). The rest constraints of HS in DEPB model are expressed as:

$$0 \le P_{\text{hs},n_{\text{hs}}}(t) \le P_{\text{hs},n_{\text{hs}},\max} \tag{67}$$

$$V_{\text{hs},n_{\text{hs}}}(t+1) = V_{\text{hs},n_{\text{hs}}}(t) + Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t) - \frac{P_{\text{hs},n_{\text{hs}}}(t)}{g\eta_{\text{hs},n_{\text{hs}}} H_{\text{hs},n_{\text{hs}}}} - V_{\text{ab},n_{\text{hs}}}(t) \tag{68}$$

$$V_{\text{hs},n_{\text{hs}}}(0) = V_{\text{hs},n_{\text{hs}}}^{w,\text{results}}(d) \tag{69}$$

$$V_{\text{hs},n_{\text{hs}}}(24) = V_{\text{hs},n_{\text{hs}}}^{w,\text{results}}(d+1) \tag{70}$$

(4) PV, WTG and transmitted power to the higher-level power grid

$$0 \le P_{\text{pv},n_{\text{pv}}}(t) \le P_{\text{pv},n_{\text{pv}}}^{\text{fore}}(t) \tag{71}$$

$$0 \le P_{\text{wtg},n_{\text{wtg}}}(t) \le P_{\text{wtg},n_{\text{wtg}}}^{\text{fore}}(t) \tag{72}$$

$$0 \le P_{\text{up}}(t) \le P_{\text{up},\max} \tag{73}$$

Similarly, the model of the MG does not include the variable P_{up}.

(5) EES

EES encompasses two states (charging and discharging). The constraints on its upper and lower limits, as well as its state of charge $SOC_{n_{\text{ees}}}$, can be expressed as follows [41]:

$$0 \le P_{\text{ees},n_{\text{ees}}}^{c}(t) \le k_{n_{\text{ees}}} P_{n_{\text{ees}},\max}^{c} \tag{74}$$

$$0 \le P_{\text{ees},n_{\text{ees}}}^{d}(t) \le (1 - k_{n_{\text{ees}}}) P_{n_{\text{ees}},\max}^{d} \tag{75}$$

$$SOC_{n_{\text{ees}}}(t+1) = SOC_{n_{\text{ees}}}(t) + (\eta_{n_{\text{ees}}}^{c} P_{\text{ees},n_{\text{ees}}}^{c}(t) - P_{\text{ees},n_{\text{ees}}}^{d}(t)/\eta_{n_{\text{ees}}}^{d})/Q_{n_{\text{ees}}} \tag{76}$$

$$SOC_{n_{\text{ees}},\min} \le SOC_{n_{\text{ees}}}(t) \le SOC_{n_{\text{ees}},\max} \tag{77}$$

where $k_{n_{\text{ees}}}$ is a binary variable, 0 and 1 represent discharging and charging state, respectively. $\eta_{n_{\text{ees}}}^{c}$ and $\eta_{n_{\text{ees}}}^{d}$ represent the charging and discharging efficiencies of EES, respectively. $Q_{n_{\text{ees}}}$ represents the energy storage capacity of EES. $SOC_{n_{\text{ees}},\min}$ and $SOC_{n_{\text{ees}},\max}$ represent the minimum and maximum $SOC_{n_{\text{ees}}}$, respectively.

(6) Fuzzy chance constraints

Similarly, v^{fore} in this section are $Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t)$, $P_{\text{pv},n_{\text{pv}}}(t)$, $P_{\text{wtg},n_{\text{wtg}}}(t)$, $P_{j,\text{load}}(t)$. The fuzzy variables of forecast error are denoted as $\Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t)$, $\Delta\tilde{P}_{\text{pv},n_{\text{pv}}}^{\text{fore}}(t)$, $\Delta\tilde{P}_{\text{wtg},n_{\text{wtg}}}^{\text{fore}}(t)$ and $\Delta\tilde{P}_{j,\text{load}}(t)$, respectively. Thus, the corresponding constraints can be expressed in the form of the following fuzzy chance constraints:

$$Cr\{\mathrm{E}(p_j(t)) + P_{j,\text{load}}(t) + \Delta\tilde{P}_{j,\text{load}}(t) = P_{ij}(t) - r_{ij} l_{ij}(t) - \sum_{k} P_{jk}(t)\} \ge \alpha \tag{78}$$

$$Cr\{\mathrm{E}(Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t)) + \Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t) - V_{\text{ab},n_{\text{hs}}}(t) \ge 0\} \ge \alpha \tag{79}$$

$$Cr\{\mathrm{E}(P_{\text{pv},n_{\text{pv}}}^{\text{fore}}(t)) + \Delta\tilde{P}_{\text{pv},n_{\text{pv}}}^{\text{fore}}(t) - P_{\text{pv},n_{\text{pv}}}(t) \ge 0\} \ge \alpha \tag{80}$$

$$Cr\{\mathrm{E}(P_{\text{wtg},n_{\text{wtg}}}^{\text{fore}}(t)) + \Delta\tilde{P}_{\text{wtg},n_{\text{wtg}}}^{\text{fore}}(t) - P_{\text{wtg},n_{\text{wtg}}}(t) \ge 0\} \ge \alpha \tag{81}$$

$$Cr\left\{ V_{\text{hs},n_{\text{hs}}}(t+1) = V_{\text{hs},n_{\text{hs}}}(t) + \mathrm{E}(Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t)) + \Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}}(t) - \frac{P_{\text{hs},n_{\text{hs}}}(t)}{g\eta_{\text{hs},n_{\text{hs}}} H_{\text{hs},n_{\text{hs}}}} - V_{\text{ab},n_{\text{hs}}}(t) \right\} \ge \alpha \tag{82}$$

Moreover, the confidence level α of the DEPB model should be higher than the confidence level α of the WEEB model. The setting principle can be seen in Section 2.2 as well.

3.4. Benefit allocation for distribution system with MMG based on shapley value

After the coordinated scheduling of MGs and DS, the total power purchased from the upper-level grid by the entire system will be reduced, and the utilization rate of renewable energy will increase. However, MGs or DS with certain regulatory capabilities will incur additional scheduling costs. Therefore, allocating benefit based solely on the actual expenses consumed by each MG or DS is unreasonable. It is necessary to achieve a reasonable benefit allocation based on the contribution of each entity to the electric energy and power balance. Therefore, the Shapley value method is employed to address the benefit allocation problem among multiple entities [42].

The Shapley value method allocates benefit according to the marginal contributions of coalition members, ensuring fairness in the allocation process. Hence, it is widely used to solve benefit allocation problems involving multiple entities. Define the total number of participants as n, and the coalition of the participants is represented as N. N_k denotes any sub-coalition that includes entity k, with s being the number of participants in this sub-coalition. Define the characteristic function of this coalition as o, where $o(N_k)$ represents the total balanced cost generated after cooperation among the entities. When the coalition is empty, the total balanced benefit is zero. The benefit allocation for DS with MMG based on the Shapley value is as follows:

$$C_k = \sum_{N_k \in N} \kappa(s) \left[o\left(N_k\right) - o\left(N_k/k\right) \right] \tag{83}$$

$$\kappa(s) = \frac{(s-1)!(n-s)!}{n!} \tag{84}$$

where C_k is the balanced benefit allocated to entity k, N_k/k is the set of entities in the coalition N_k excluding k, and $\kappa(s)$ is the probability of the occurrence of N_k in the coalition N.

4. Solution method

4.1. Crisp equivalent class transformation method of fuzzy chance constraints

Since the above model is a stochastic optimization problem with fuzzy chance constraints, it is difficult to solve it directly. Thus, the crisp equivalent class transformation method of fuzzy chance constraints is employed to solve this problem, which can transform the original complex uncertain optimization problems into deterministic LP and MISOCP problems. If the constraints involve fuzzy variables χ, they take the following form:

$$g(x,\chi) = h_0(x) + h_1(x)\chi_1 + h_2(x)\chi_2 + \cdots + h_K(x)\chi_K \tag{85}$$

where χ_n represents the relevant fuzzy parameters of the fuzzy membership function $\mu(\chi)$, which decides the shape of $\mu(\chi)$. The fuzzy parameters of the forecasted errors for generation and demand within each scheduling interval can be represented using trapezoidal membership functions:

$$\mu(\chi) = \begin{cases} \dfrac{\chi - \chi_1}{\chi_2 - \chi_1}, & \chi_1 < \chi < \chi_2 \\ 1, & \chi_2 < \chi < \chi_3 \\ \dfrac{\chi - \chi_4}{\chi_4 - \chi_3}, & \chi_3 < \chi < \chi_4 \\ 0, & \text{Otherwise} \end{cases} \tag{86}$$

The use of trapezoidal fuzzy parameter equivalence classes for fuzzy chance constraints enables decision-making solutions that balance risks and costs. Moreover, h_k is defined as:

$$h_k^+ = \begin{cases} h_k(x), & h_k(x) \geq 0 \\ 0, & h_k(x) < 0 \end{cases} \tag{87}$$

$$h_k^- = \begin{cases} 0, & h_k(x) \geq 0 \\ -h_k(x), & h_k(x) < 0 \end{cases} \tag{88}$$

where $k = 1, 2, \ldots, K, K \in R$. When $\alpha \geq 0.5$ and χ conforms to the trapezoidal membership function, constraints $Cr\{g(x,\chi) \leq 0\} \geq \alpha$ can be transformed into crisp equivalents:

$$(2-2\alpha)\sum_{k=1}^{K}[\chi_3 h_k^+(x) - \chi_2 h_k^-(x)] + (2\alpha - 1)\sum_{k=1}^{K}[\chi_4 h_k^+(x) - \chi_1 h_k^-(x)] + h_0(x) \leq 0 \tag{89}$$

Taking Eq. (33) as an example, the constraint function can be expressed as:

$$g(V_{\text{ab},n_{\text{hs}}}, \Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}}) = \text{E}(Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(w)) + \Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}}(w) - V_{ab,n_{hs}}(w) = h_0(V_{\text{ab},n_{\text{hs}}}) + h_1(V_{\text{ab},n_{\text{hs}}})\Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}} \tag{90}$$

$$\begin{cases} h_0(V_{\text{ab},n_{\text{hs}}}) = \text{E}(Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(w)) - V_{\text{ab},n_{\text{hs}}}(w) \\ h_1(V_{\text{ab},n_{\text{hs}}}) = 1 \end{cases} \tag{91}$$

According to Eqs. (87) and (88), h_1^+ and h_1^- are equal to 1 and 0, respectively. Define the trapezoidal fuzzy parameters of $\Delta\tilde{Q}_{\text{hs},n_{\text{hs}}}^{\text{fore}}$ as $\Delta\tilde{Q}_{\text{hs},n_{\text{hs}},1}^{\text{fore}}$, $\Delta\tilde{Q}_{\text{hs},n_{\text{hs}},2}^{\text{fore}}$, $\Delta\tilde{Q}_{\text{hs},n_{\text{hs}},3}^{\text{fore}}$ and $\Delta\tilde{Q}_{\text{hs},n_{\text{hs}},4}^{\text{fore}}$. The crisp equivalents of Eq. (33) can be expressed as:

$$\text{E}(Q_{\text{hs},n_{\text{hs}}}^{\text{fore}}(w)) + (2-2\alpha)\Delta Q_{\text{hs},n_{\text{hs}},2}^{\text{fore}}(w) + (2\alpha - 1)\Delta Q_{\text{hs},n_{\text{hs}},1}^{\text{fore}}(w) - Q_{\text{hs},n_{\text{hs}}}(w) \geq 0 \tag{92}$$

Similarly, the fuzzy chance constraints of proposed model Eqs. (34)–(38), (50)–(55), (78)–(82) can be transformed into:

(1) MEEB model

$$\mathrm{E}(E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(w)) + (2-2\alpha)\Delta E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}},2}(w) + (2\alpha-1)\Delta E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}},1}(w) - E_{\mathrm{pv},n_{\mathrm{pv}}}(w) \geq 0 \tag{93}$$

$$\mathrm{E}(E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(w)) + (2-2\alpha)\Delta E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}},2}(w) + (2\alpha-1)\Delta E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}},1}(w) - E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w) \geq 0 \tag{94}$$

$$\begin{aligned} V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(w+\Delta w) =& V^{m}_{\mathrm{hs},n_{\mathrm{hs}}}(w) + \mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(w)) + (2-2\alpha)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},2}(w) + (2\alpha-1)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},1}(w) \\ &- \frac{E_{\mathrm{hs},n_{\mathrm{hs}}}(w)}{g\eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}}} - V_{\mathrm{ab},n_{\mathrm{hs}}}(w) \end{aligned} \tag{95}$$

$$\begin{aligned} &E_{\mathrm{up}}(w) + \sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}} \left[E_{\mathrm{h2p},n_{\mathrm{h}}}(w) - E_{\mathrm{p2h},n_{\mathrm{h}}}(w) - E_{\mathrm{h},n_{\mathrm{h}}}(w)\right] + \sum_{n_{\mathrm{pv}}\in N_{\mathrm{pv}}} E_{\mathrm{pv},n_{\mathrm{pv}}}(w) + \sum_{n_{\mathrm{wtg}}\in N_{\mathrm{wtg}}} E_{\mathrm{wtg},n_{\mathrm{wtg}}}(w) + \\ &\sum_{n_{\mathrm{hs}}\in N_{\mathrm{hs}}} E_{\mathrm{hs},n_{\mathrm{hs}}}(w) = \mathrm{E}(E_{\mathrm{load}}(w)) + (2-2\alpha)\Delta E_{\mathrm{load},2}(w) + (2\alpha-1)\Delta E_{\mathrm{load},1}(w) + \Delta E_{\mathrm{loss,week}}(w) \end{aligned} \tag{96}$$

$$\begin{aligned} &[\sum_{d\in \boldsymbol{D}} P_{\mathrm{up,max}}\Delta d - E_{\mathrm{up}}(w)] + \min[g\eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}} V_{\mathrm{hs},n_{\mathrm{hs}}}(w+1), \sum_{d\in \boldsymbol{D}} P_{\mathrm{hs},n_{\mathrm{hs}},\max}\Delta d] \geq \\ &\rho_{\mathrm{load},1}\left(\mathrm{E}(E_{\mathrm{load}}(w)) + (2-2\alpha)\Delta E_{\mathrm{load},3}(w) + (2\alpha-1)\Delta E_{\mathrm{load},4}(w)\right) \end{aligned} \tag{97}$$

(2) WEEB model

$$\mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(d)) + (2-2\alpha)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},2}(d) + (2\alpha-1)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},1}(d) - Q_{\mathrm{hs},n_{\mathrm{hs}}}(d) \geq 0 \tag{98}$$

$$\mathrm{E}(E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(d)) + (2-2\alpha)\Delta E^{\mathrm{fore}}_{pv,n_{pv},2}(d) + (2\alpha-1)\Delta E^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}},1}(d) - E_{\mathrm{pv},n_{\mathrm{pv}}}(d) \geq 0 \tag{99}$$

$$\mathrm{E}(E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(d)) + (2-2\alpha)\Delta E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}},2}(d) + (2\alpha-1)\Delta E^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}},1}(d) - E_{\mathrm{wtg},n_{\mathrm{wtg}}}(d) \geq 0 \tag{100}$$

$$\begin{aligned} V_{\mathrm{hs},n_{\mathrm{hs}}}(d+1) =& V_{\mathrm{hs},n_{\mathrm{hs}}}(d) + \mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(d)) + (2-2\alpha)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},2}(d) \\ &+ (2\alpha-1)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},1}(d) - \frac{E_{\mathrm{hs},n_{\mathrm{hs}}}(d)}{g\eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}}} - V_{\mathrm{ab},n_{\mathrm{hs}}}(d) \end{aligned} \tag{101}$$

$$\begin{aligned} &E_{\mathrm{up}}(d) + \sum_{n_{\mathrm{h}}\in N_{\mathrm{h}}} \left[E_{\mathrm{h2p},n_{\mathrm{h}}}(d) - E_{\mathrm{p2h},n_{\mathrm{h}}}(d) - E_{\mathrm{h},n_{\mathrm{h}}}(d)\right] + \sum_{n_{\mathrm{pv}}\in N_{\mathrm{pv}}} E_{\mathrm{pv},n_{\mathrm{pv}}}(d) + \sum_{n_{\mathrm{wtg}}\in N_{\mathrm{wtg}}} E_{\mathrm{wtg},n_{\mathrm{wtg}}}(d) + \\ &\sum_{n_{\mathrm{hs}}\in N_{\mathrm{hs}}} E_{\mathrm{hs},n_{\mathrm{hs}}}(d) = \mathrm{E}(E_{\mathrm{load}}(d)) + (2-2\alpha)\Delta E_{\mathrm{load},3}(d) + (2\alpha-1)\Delta E_{\mathrm{load},4}(d) + \Delta E_{\mathrm{loss,day}}(d) \end{aligned} \tag{102}$$

$$\begin{aligned} &[\sum_{t\in \boldsymbol{T}} P_{\mathrm{up,max}}\Delta t - E_{\mathrm{up}}(d)] + \min[g\eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}} V_{\mathrm{hs},n_{\mathrm{hs}}}(d+1), \sum_{t\in \boldsymbol{T}} P_{\mathrm{hs},n_{\mathrm{hs}},\max}\Delta t] \geq \\ &\rho_{\mathrm{load},2}\left(\mathrm{E}(E_{\mathrm{load}}(d)) + (2-2\alpha)\Delta E_{\mathrm{load},3}(d) + (2\alpha-1)\Delta E_{\mathrm{load},4}(d)\right) \end{aligned} \tag{103}$$

(3) DEPB model

$$p_j(t) + \mathrm{E}(P_{j,\mathrm{load}}(t)) + (2-2\alpha)\Delta P_{j,\mathrm{load},2}(t) + (2\alpha-1)\Delta P_{j,\mathrm{load},1}(t) = P_{ij}(t) - r_{ij}l_{ij}(t) - \sum_k P_{jk}(t) \tag{104}$$

$$\mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(t)) + (2-2\alpha)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},2}(t) + (2\alpha-1)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},1}(t) - Q_{\mathrm{hs}.n_{\mathrm{hs}}}(t) \geq 0 \tag{105}$$

$$\mathrm{E}(P^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}}}(t)) + (2-2\alpha)\Delta P^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}},2}(t) + (2\alpha-1)\Delta P^{\mathrm{fore}}_{\mathrm{pv},n_{\mathrm{pv}},1}(t) - P_{\mathrm{pv},n_{\mathrm{pv}}}(t) \geq 0 \tag{106}$$

$$\mathrm{E}(P^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}}}(t)) + (2-2\alpha)\Delta P^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}},2}(t) + (2\alpha-1)\Delta P^{\mathrm{fore}}_{\mathrm{wtg},n_{\mathrm{wtg}},1}(t) - P_{\mathrm{wtg},n_{\mathrm{wtg}}}(t) \geq 0 \tag{107}$$

$$\begin{aligned} V_{\mathrm{hs},n_{\mathrm{hs}}}(t+1) =& V_{\mathrm{hs},n_{\mathrm{hs}}}(t) + \mathrm{E}(Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}}}(t)) + (2-2\alpha)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},2}(t) + (2\alpha-1)\Delta Q^{\mathrm{fore}}_{\mathrm{hs},n_{\mathrm{hs}},1}(t) \\ &- \frac{P_{\mathrm{hs},n_{\mathrm{hs}}}(t)}{g\eta_{\mathrm{hs},n_{\mathrm{hs}}} H_{\mathrm{hs},n_{\mathrm{hs}}}} - V_{\mathrm{ab},n_{\mathrm{hs}}}(t) \end{aligned} \tag{108}$$

Through this transformation, the proposed model can be transformed into a certain model which can be solved through commercial solver.

4.2. ADMM algorithm

The ADMM algorithm is widely employed in addressing distributed optimization problems due to its favorable convergence properties and simple formalism. The ADMM algorithm can be utilized to solve the following optimization problem:

$$\begin{cases} \min f(x) + g(z) \\ \text{s.t. } x \in \boldsymbol{X}, z \in \boldsymbol{Z} \\ \quad Ax + Bz = c \end{cases} \tag{109}$$

where, $\boldsymbol{X}$ and $\boldsymbol{Z}$ are convex sets, and the optimization model in Eq. (109) is convex. f and g respectively represent the objective functions of the sub-optimization problems. For the multi-microgrid coordinated scheduling problem addressed in this paper, the sub-optimization problems can be expressed as the optimization scheduling problems for each region (the DS and the MG), where x and z in Eq. (109) represent the consistency variables for each region. In order to conduct distributed optimization on the temporal scale of daily interactions within MMG DS discussed earlier, the decoupling approach proposed in Section 3.3 is initially employed to decouple the variables of the DS and the MG. Subsequently, EMS-MG and EMS-DS formulate sub-optimization problems, and the iterative computations facilitated by ADMM algorithm are employed to obtain the global optimum.

For the sub-optimization problem f_A of area A (the DS or the MG), the iterative format of the consistency variable X_A and the Lagrange multiplier u_A in ADMM can be expressed as [28,43]:

$$X_A^{iter+1} = \arg\min f_A(X_A) + \frac{\rho^{iter}}{2} \sum_{j\in\Omega_A} \left\| X_{A,j} - u_{A,j}^{iter} \right\|_2^2 \tag{110}$$

$$u_{A,j}^{iter+1} = u_{A,j}^{iter} + X_{B,j}^{iter+1} - \frac{X_{A,j}^{iter} + X_{B,j}^{iter}}{2} \tag{111}$$

where iter represents the iteration count, ρ^{iter} represents the iteration step size in iterth iteration, Ω_A represents the collection of boundary nodes connected to area A, area B is the DS or the MG which is connected to area A. Substitute the consistency variables with their respective counterparts:

$$X_A^{iter+1} = \arg\min f_A(X_A) + \frac{\rho^{iter}}{2} \sum_{t=1}^{24} \sum_{i\in A, j\in\Omega_A} \begin{array}{l} [(P_{ij'}(t) - \hat{P}_{ij'}^{iter}(t))^2 + \\ (Q_{ij'}(t) - \hat{Q}_{ij'}^{iter}(t))^2 + \\ (l_{ij'}(t) - \hat{l}_{ij'}^{iter}(t))^2] \end{array} \tag{112}$$

where $\hat{P}_{ij'}^{iter}$, $\hat{Q}_{ij'}^{iter}$ and $\hat{l}_{ij'}^{iter}$ represent the reference values for the consistency variables of linking lines in the iterth iteration, respectively, which can be calculated as:

$$\begin{cases} \hat{P}_{ij'}^{iter+1} = \hat{P}_{ij'}^{iter} - (P_{ij'}{}^{iter+1} + P_{ji'}{}^{iter+1})/2 \\ \hat{Q}_{ij'}^{iter+1} = \hat{Q}_{ij'}^{iter} - (Q_{ij'}{}^{iter+1} + Q_{ji'}{}^{iter+1})/2 \\ \hat{l}_{ij'}^{iter+1} = \hat{l}_{ij'}^{t} - (l_{ij'}{}^{iter+1} + l_{ji'}{}^{iter+1})/2 \end{cases} \tag{113}$$

The convergence characteristics of the algorithm are described by the primal residuals $r_{ij'}^{iter}$ and dual residuals $s_{ij'}^{iter}$ at each iteration:

$$\begin{cases} r_{ij'}^{iter} = \left\| X_{ij'}^{iter} - u_{ij'}^{iter} \right\| \\ s_{ij'}^{iter} = \left\| X_{ij'}^{iter} - X_{ij'}^{iter-1} \right\| \end{cases} \tag{114}$$

The termination criterion for the algorithm is established such that it concludes when either the primal residual or the dual residual descends below a threshold, denoted as $10^{-4}\sqrt{N_{\mathrm{node}}}$, where N_{node} represents the total number of nodes in the system.

Due to the influence of the convergence of the coupling variable on ρ^{iter}, the traditional ADMM assumes a constant step size, and consequently, its convergence speed is significantly affected by the chosen value of ρ^{iter}. In order to enhance the convergence characteristics of the algorithm, an adaptive step size adjustment mechanism is introduced to dynamically regulate ρ^{iter} [28]:

$$\rho^{iter+1} = \begin{cases} \rho^{iter} / \left(1 + \lg\left(s^{iter}/r^{iter}\right)\right), & s^{iter} < 0.1 r^{iter} \\ \rho^{iter}\left(1 + \lg\left(r^{iter}/s^{iter}\right)\right), & s^{iter} > 0.1 r^{iter} \\ \rho^{iter}, & \text{otherwise} \end{cases} \tag{115}$$

By introducing this mechanism, when there is a significant disparity between s^{iter} and r^{iter}, the adjustment of ρ^{iter} can be employed to alter the weighting of the consistency cost, thereby avoiding oscillations in the objective function and expediting the convergence speed. The iterative process of the variable step-size ADMM algorithm (SADMM) is illustrated in Fig. 5, and the specific steps are as follows:

(1) Initialize the iteration operators and consistency variable values. Set the termination criterions.

(2) The EMS-DS updates and stores Lagrange multipliers based on Eq. (113), and distribute the Lagrange multipliers to the EMS-MG.

(3) The EMS-DS calculates both primal and dual residuals, evaluates whether the termination criteria are satisfied. If the criteria are met, the iteration concludes, and the optimization results are output; otherwise, the algorithm proceeds to step (4).

(4) Each EMS-MG and EMS-DS receives Lagrange multiplier information, formulates sub-optimization problems, and engages in distributed optimization. Set $iter = iter + 1$ and update ρ^{iter}.

(5) The EMS-MG and EMS-DS resolve the sub-problems and acquire consistency variables. Subsequently, the EMS-MG consolidates the consistency variables, transmits it to the EMS-DS, and proceeds to step (2).

The computational flowchart for the multi-timescale coordinated scheduling method for DS with multi-microgrid is depicted in Fig. 6. The proposed method encompasses three stages: monthly electric energy balance, weekly electric energy balance, and daily electric power balance. Initially, the monthly balancing process is executed by inputting the forecasted source-load electric energy data for the upcoming month. The MEEB model is then solved to determine the weekly initial and final states of the HS' reservoir and HSS. This iterative procedure continues on a rolling basis until the end of the respective month. Subsequently, the weekly balancing stage commences with the input of forecasted source-load electric energy data for the upcoming week. The WEEB model is solved to ascertain the daily initial and final states of the HS' reservoir and HSS, with this iterative process persisting on a rolling basis until the conclusion of the week. Finally, hourly day-ahead source-load power forecast data is incorporated, and the DEPB model is addressed through the application of the SADMM algorithm. This yields the day-ahead scheduling strategy for the MMG system. The scheduling costs for each entity are calculated using the cost allocation method described in Section 3.4.

5. Results and discussion

In this chapter, the effectiveness and the applicability of the proposed scheduling strategy are first validated through a distribution system with two MGs under various source-load scenarios. Specifically, the daily average curtailment rate and daily average cost of different methods under

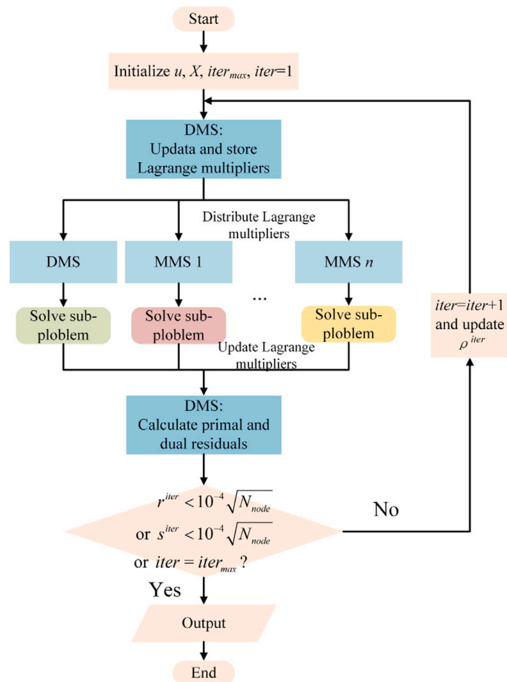


Fig. 5. Flowchart of SADMM algorithm.

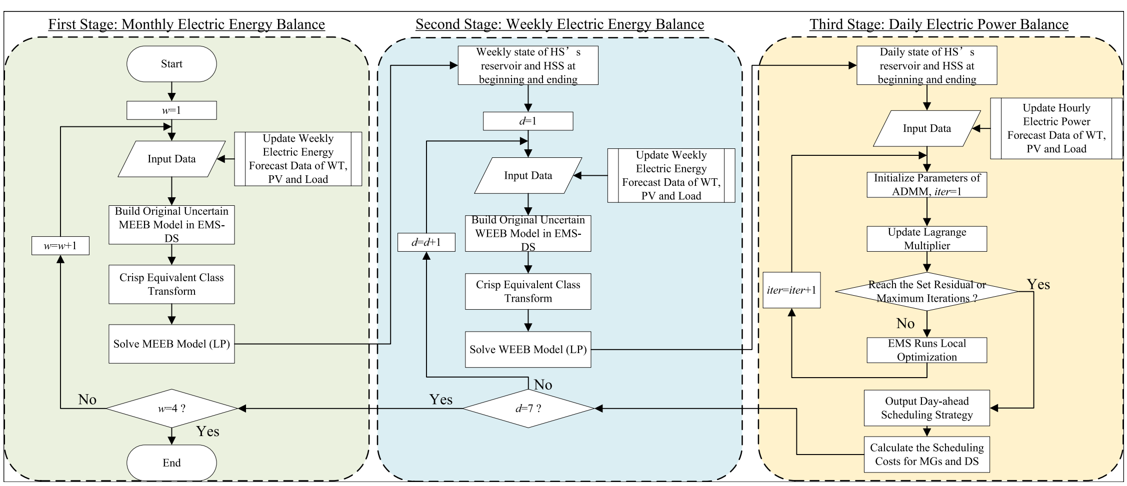


Fig. 6. Flowchart of proposed multi-timescale coordinated scheduling method.

various seasonal scenarios are compared. Secondly, to verify the economic efficiency and renewable energy accommodation performance of the proposed coordinated dispatching strategy across different time scales, the scheduling results of different dispatching strategies at three time scales: monthly, weekly, and daily are compared. Furthermore, to validate the feasibility of the proposed benefits allocation method, a comparative analysis of the benefits of MGs and DS under both alliance and non-alliance scenarios is conducted. Finally, to verify the optimality and computational efficiency of the proposed method, the solution time and optimization results of the traditional centralized solving algorithm and the traditional distributed solving algorithm are compared.

5.1. Case study setting

A DS with multiple microgrids, based on the IEEE 15-node test case [44], is employed to validate the effectiveness of the proposed methodology. The system's topology is illustrated in Fig. 7 and includes a PV (PV1), a WTG (WTG1), a HS (HS1), two EES (EES1, EES2), a HS (HS1), and two MGs (MG1 and MG2). Specifically, MG1 adopts the topology structure of the IEEE 33-node test case [45], incorporating a HS (HS2) connected to node 22, a HS (HS2) connected to node 11, an EES unit (EES3) connected to node 22, a WTG (WTG2) connected to node 22, and a PV (PV2) connected to node 13. On the other hand, MG2 adopts the topology structure of the IEEE 51-node test case [46], featuring an EES (EES4) connected to node 22, a WTG (WTG3) connected to node 42, and a PV (PV3) connected to node 22. The parameters of each resource are detailed in Tables A.1–A.3. Load forecast data, as well as electric energy and power forecast data for PV, WTG, and HS, have been proportionally adapted based on publicly available datasets from the European Union and its neighboring regions [47]. The proposed method is developed by using MATLAB R2023a with

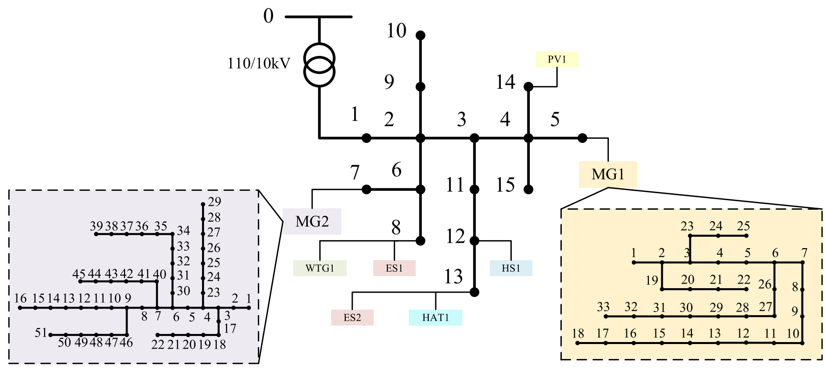


Fig. 7. Topology of MMG system.

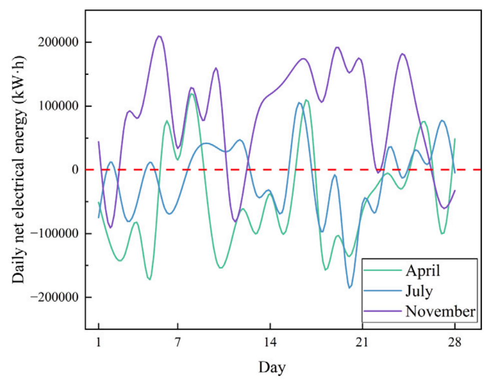


Fig. 8. Daily net electrical energy load of April, July and November.

Table 1
Comparison of daily average system scheduling costs and daily average power curtailment rates for M1–M4.

		M1	M2	M3	M4
April	Daily average cost	17.2572	14.5249	10.3861	9.3309
	Daily average power curtailment rate	6.6209	5.7483	5.6327	5.0232
July	Daily average cost	8.0287	5.0101	2.7048	1.6348
	Daily average power curtailment rate	5.6290	4.7037	4.9389	4.2631
November	Daily average cost	7.3911	7.1188	7.0757	6.8616
	Daily average power curtailment rate	4.4347	2.9996	1.5810	1.3247

the commercial optimization solver GUROBI. All simulations are performed on a PC platform with 16 GB RAM and Intel Core i7-12700H CPUs (2.70 GHz).

5.2. Validation of the effectiveness under various scenarios

To validate the effectiveness of the proposed multi-energy storage system, three benchmark methods and proposed method are established:

M1: Scheduling method considering the EES [48]

M2: Scheduling method considering the multi-energy storage system composed of HS and EES [41]

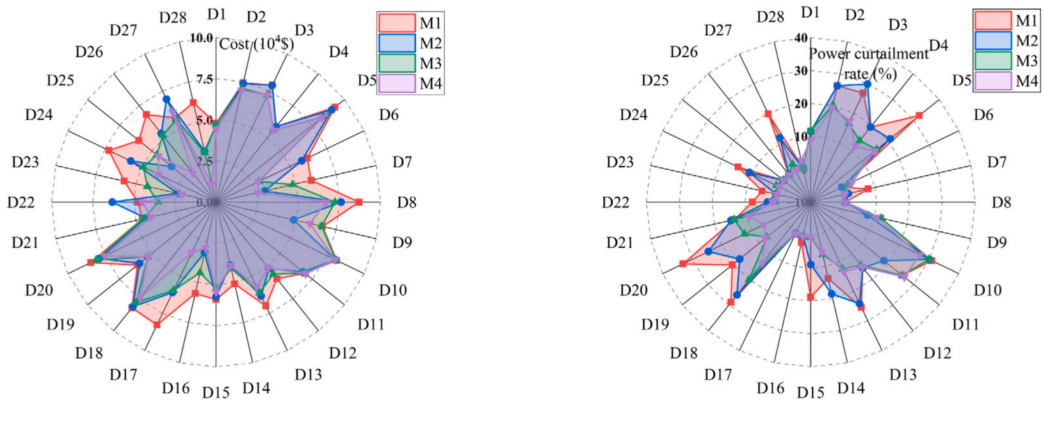
M3: Scheduling method considering the multi-energy storage system composed of HSS and EES [49]

M4: Proposed method

To validate the adaptability of the proposed method in different scenarios, the source-load forecast data from the 1st to the 28th of April, July, and November of the region as three typical scenarios for the scheduling methods M1–M4 are selected. The daily net electrical energy load curves for the three typical months are shown in Fig. 8. In April, the region experiences high wind power generation, resulting in an overall negative net load, with a daily average net load of −45,671 kWh. In July, although photovoltaic generation is high, the overall load level is relatively high, with a daily average net load of −14,399 kWh. In November, renewable energy generation decreases, leading to a daily average net load of 84,476 kWh.

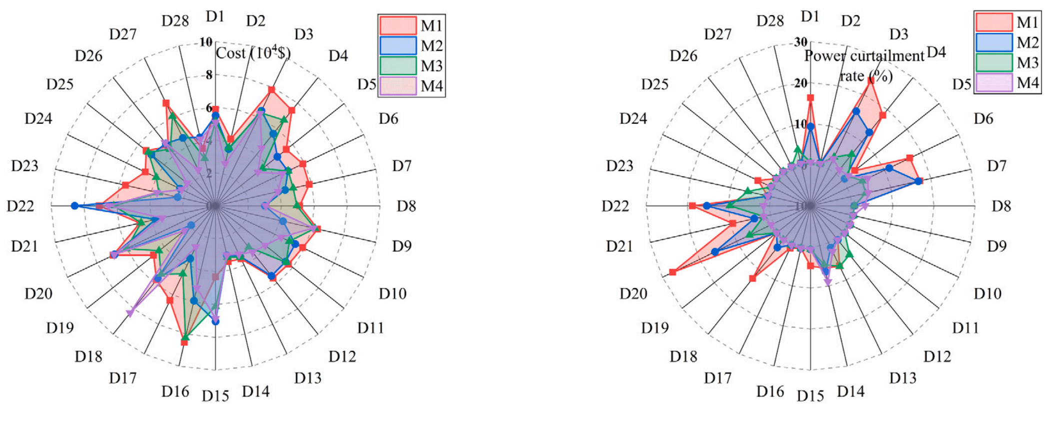
The scheduling of M1–M4 was applied to the above-mentioned scenarios, and the daily curtailment rates and daily operational costs for each month are shown in Figs. 9–10. Additionally, the daily average scheduling costs and daily average curtailment rates for M1–M4 are presented in Table 1.

As shown in Fig. 9, in April, the renewable energy generation is high, and all methods exhibit a high curtailment rate. The M1, which only considers the regulation of EES within a day, has the poorest regulation ability and the highest scheduling cost. The M2 and M3, which take into account the regulation capabilities of HS and HSS during daily and weekly periods, respectively, enhance operational flexibility. However,



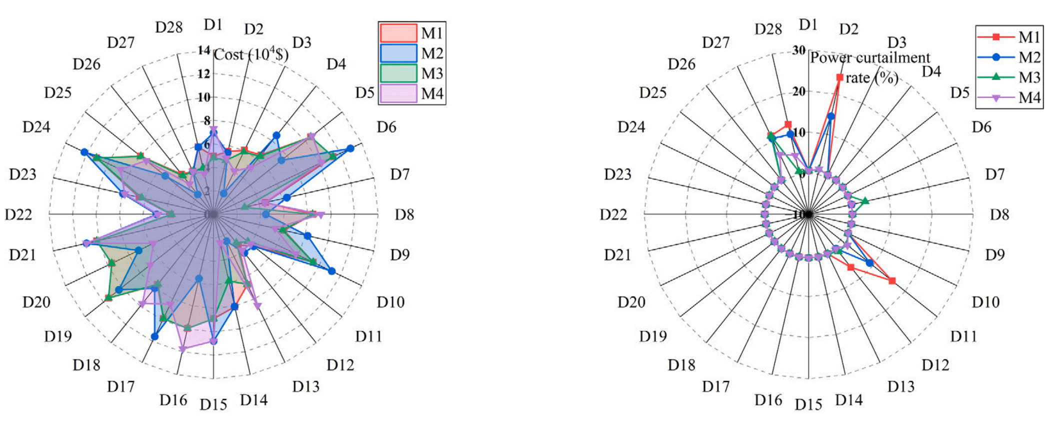
(a) (b)

Fig. 9. Comparison of (a) costs (b) power curtailment rate for M1–M4 in April.



(a) (b)

Fig. 10. Comparison of (a) costs (b) power curtailment rate for M1–M4 in July.



(a) (b)

Fig. 11. Comparison of (a) costs (b) power curtailment rate for M1–M4 in November.

since hydropower generation is significantly influenced by water availability, its regulation ability is weaker than that of HSS. Therefore, overall, the M3 results in lower daily average scheduling costs and curtailment rates compared to M2. The M4, utilizing a multi-energy storage system, effectively considers regulation capabilities across three timescales: monthly, weekly, and daily, leading to better operational flexibility. As a result, M4 demonstrates lower daily average scheduling costs and curtailment rates than the benchmark methods.

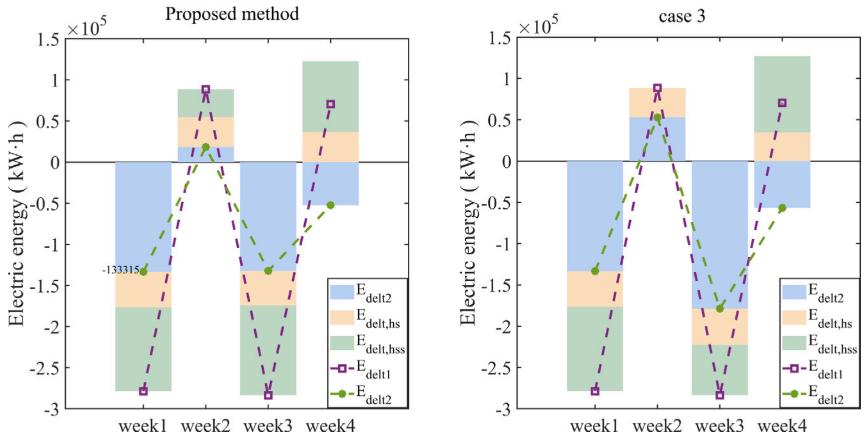


Fig. 12. Net load electric energy at monthly time scale.

Similarly, in July, when renewable energy generation and load are relatively balanced, the proposed method significantly reduces curtailment rates and scheduling costs, which is shown in Fig. 10. In November, where net load becomes predominantly positive, the overall curtailment rate is relatively low, with only occasional high wind generation days causing the net load to be negative. Except for M1, which lacks the ability to shift daytime generation, M2–M4 can transfer the daytime generation, thus reducing curtailment rates on days with high wind generation, which is shown in Fig. 11. Overall, as shown in Table 1, the proposed method provides economically feasible scheduling strategies in all three typical monthly scenarios, improving the integration of renewable energy.

5.3. The impact of different scheduling strategies across multiple time scales

In order to assess the effectiveness of the proposed method in temporal dimension, three comparative cases are established:

case 1: Both HSS and HS employ a weekly operation mode, utilizing the WEEB and DEPB models for resolution.

case 2: Both HSS and HS employ a daily operation mode, utilizing the DEPB models for resolution.

case 3: In contrast to the proposed method, the method does not employ rolling optimization.

The comparison between case1 and the proposed method is employed to analyze the advantage of the method in considering the inter-week electricity regulation capabilities of HSS and HS; the comparison between case2 and the proposed method is employed to analyze the advantage of the method in considering both inter-week and intra-day electricity regulation capabilities of HSS and HS; the comparison between case3 and the proposed method is employed to analyze the advantage of the method in employing rolling optimization, and for the sake of brevity, this comparison is confined to the monthly time scale.

A. At month time scale

Due to the absence of intra-week electric energy regulation capabilities in the flexibility resources considered in case 1 and case 2, they do not exhibit the ability to smooth the intra-month electricity demand fluctuations. Here, a comparison is made between two methods: one employing rolling optimization (proposed method) and the other not (case 3). Fig. 12 depicts the monthly net load energy (E_{delt1}) and equivalent net load energy (E_{delt2}) after monthly electric energy regulation using HS ($E_{\mathrm{delt,hs}}$) and HSS ($E_{\mathrm{delt,hss}}$) of two methods. It can be observed that in week1 and week3, the system's net load energy becomes negative, indicating that the renewable energy generation exceeds the electricity consumption. Conversely, in week2 and week4, the system's net load energy becomes positive, signifying that the renewable energy generation is less than the electricity consumption. Fig. 13 illustrates that the peak-to-trough difference in net load energy is significantly reduced through the electricity regulation provided by HS and HSS for both two methods. However, with the implementation of rolling optimization, the fluctuations in E_{delt2} during the latter three weeks are reduced. Therefore, it is inferred that rolling optimization has the capability to mitigate the impact of prediction errors on dispatch strategies.

Fig. 13 contrasts the results of HS's reservoir and HSS dispatch strategies for two methods. It can be observed that the utilization of HSS with rolling optimization (proposed method) is more frequent (due to the lower cost of utilizing water, and both reservoirs having the same capacity, resulting in the full utilization of hydropower in both cases). Combining the analysis of the electricity deficit, it is evident that discharging is required in week2 and week4 when the net load energy becomes positive, and charging is necessary in week1 and week3 when the net load energy becomes negative. Due to prediction errors, it is noted that without rolling optimization (case 3), there is underutilization of HSS in week1 and week3, leading to less efficient charging and discharging strategies compared to the results obtained with rolling optimization.

The comparison of costs and curtailment rates among case1, case2 and proposed methods at the monthly time scale is illustrated in Fig. 14. The proposed method exhibits the minimum curtailment rate in week1 and week3 and the minimum dispatch cost in week2 and week4. The average cost of case1, case2 and proposed method are 44 863.7$, 48 259.8$, and 41 644.9$ respectively. And the average curtailment rate of case1, case2 and proposed method are 7.02%, 8.59%, and 4.07% respectively. Overall, the proposed method demonstrates the lowest average cost and curtailment rate. However, in week2 and week4, the curtailment rate is slightly higher than that of case 1. This is attributed to the greater intra-week flexibility adjustment capacity in case 1 compared to the proposed method. In week1, the higher cost of the proposed method compared to case 1 is due to the need for the proposed method to dispatch more resources to achieve a lower curtailment rate, resulting in a slightly higher cost.

B. At week time scale

Similarly, due to the absence of intra-day electric energy regulation capabilities in the flexibility resources considered in case 2, a comparison is made between case1 and proposed method. Fig. 15 depicts the daily net load energy (E_{delt1}) and equivalent net load energy (E_{delt2}) after weekly electric energy regulation using HS ($E_{\mathrm{delt,hs}}$) and HSS ($E_{\mathrm{delt,hss}}$) of two methods in week1. It can be observed that in day1, day3, day4, day6 and

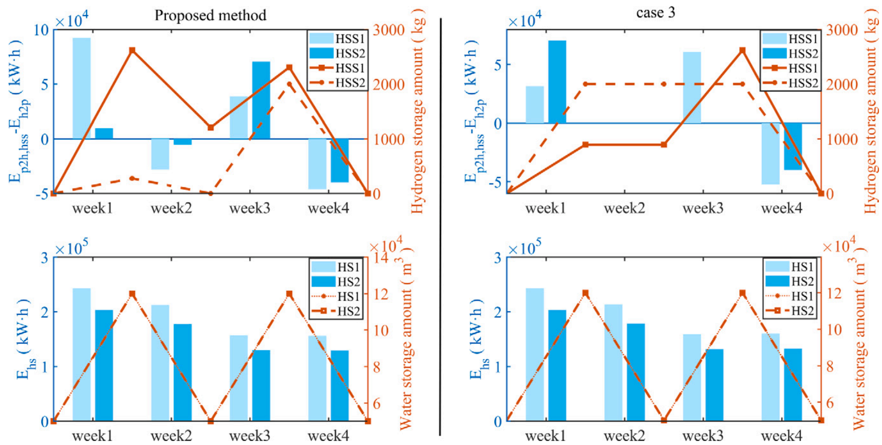
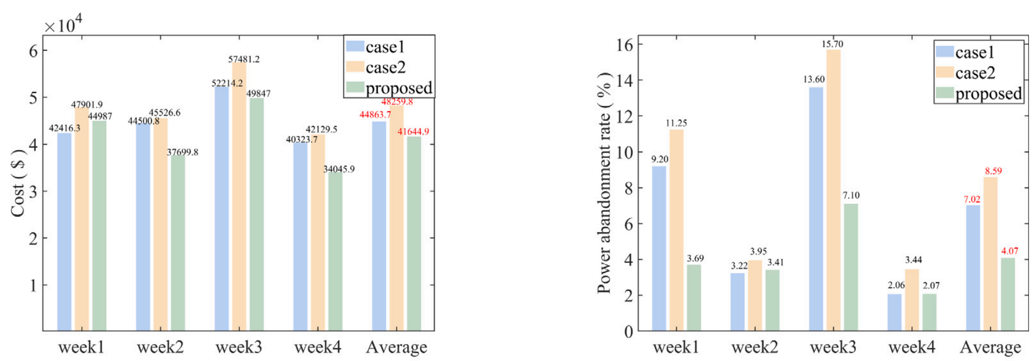


Fig. 13. HS's reservoir and HSS dispatch strategies.



(a) (b)

Fig. 14. Comparison of (a) costs (b) power curtailment rate for three methods.

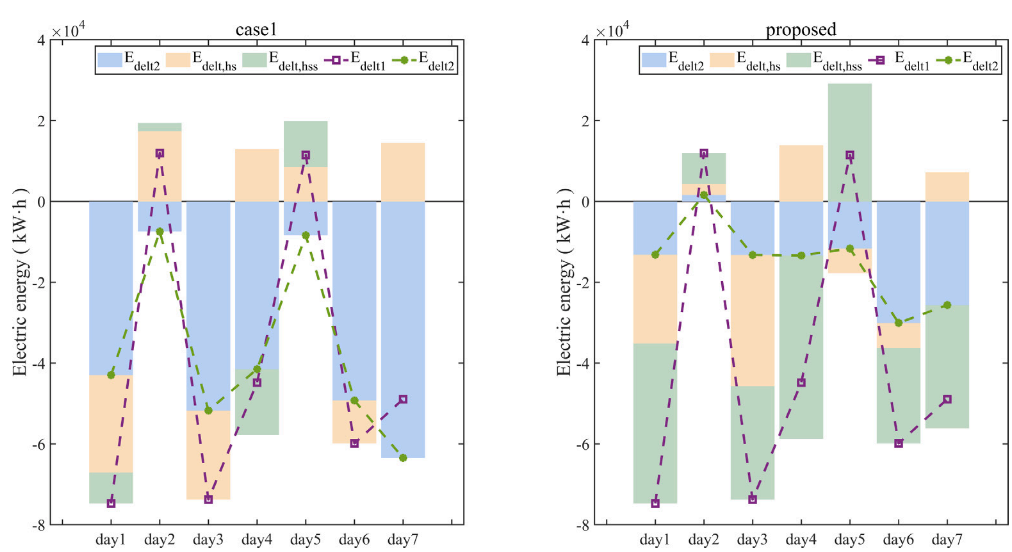


Fig. 15. Net load electric energy at weekly time scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

day7, the system's net load energy becomes negative, and in day2 and day5, the system's net load energy becomes positive. Fig. 15 illustrates that, in comparison to case 1, the proposed method exhibits a significant reduction in the peak-to-trough difference in the equivalent net load curve (green line). And the HS's reservoir strategies for case1 and proposed method in week1 are shown in Fig. 16. The proposed method demonstrates a higher water storage capacity within the initial week, whereas the reservoir capacity in case 1 is considered to remain consistent at the beginning and end of each week. The reservoir operation strategy of proposed method is characterized by enhanced flexibility.

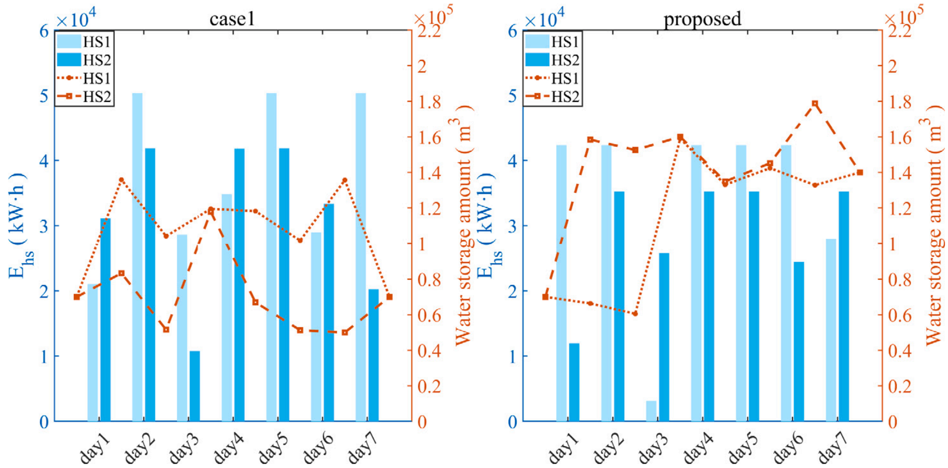
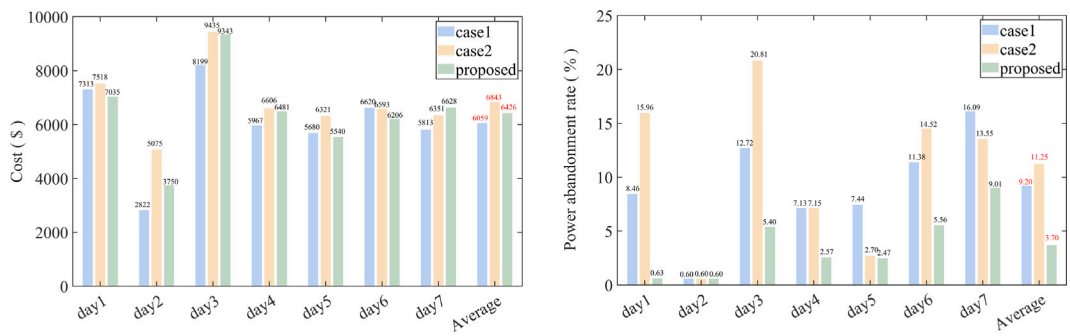


Fig. 16. HS's reservoir dispatch strategies.



(a) (b)

Fig. 17. Comparison of (a) costs (b) power curtailment rate for three methods.

The comparison of costs and curtailment rates among the three methods in week1 is illustrated in Fig. 17. The average cost of case1, case2 and proposed method are 6059.5$, 6843.1$, and 6426.7$ respectively. And the average curtailment rate of case1, case2 and proposed method are 9.20%, 11.25%, and 3.70% respectively. The proposed method exhibits the minimum curtailment rate among three methods. In addition, the cost of proposed method is at its minimum on day1, day5 and day6, slightly higher on day2, day3 and day4 compared to the cost in case 1, but lower than the cost in case 2. Overall, due to the increased flexibility required by the proposed method to mitigate curtailment rates, the associated costs for week1 are marginally higher than those in case 1, while Case 2 incurs higher curtailment costs due to its elevated curtailment rates.

C. At day time scale

At daily time scale, an initial analysis is conducted on the state of charge (SOC_{all}) and output of EES (P_{ees}), as depicted in Fig. 18. The trends in SOC variation for the three methods are generally consistent, primarily involving charging during periods of abundant renewable energy generation throughout the day and discharging during the remaining time intervals. The SOC variation ranges for case 1, case 2, and the proposed method are 0.17–0.72, 0.28–0.61, and 0.02–0.82, respectively. The proposed approach exhibits a higher energy storage utilization efficiency.

Subsequently, the flexibility resource demand ($P_{\mathrm{fle,n}}$), internal flexibility resource output (P_{fle}), and interconnection line power (P_{1-2} and P_{1-3}) as well as the power from the higher-level grid (P_{up}) are demonstrated in the spatial dimension for each microgrid and the DS, as shown in Fig. 19. It is evident that the proposed method facilitates a greater transmission of power through interconnection lines, thereby enabling a more comprehensive utilization of flexibility resources across different regions to address deficiencies in their own flexibility resources. In contrast, case 1 and case 2, due to ineffective utilization of flexibility resources, result in significant curtailment of power.

Furthermore, the cost and electric energy curtailment of day3 at various confidence levels are analyzed, as depicted in Fig. 20. In order to achieve electric energy and power balance at higher confidence levels, the system necessitates increased expenditure. Consequently, with the elevation of confidence levels, both curtailment rates and scheduling costs exhibit an upward trend.

5.4. Economic analysis of proposed strategy

The costs and benefits of the DS and MGs are analyzed in this section. The costs of the DS and MGs mainly arise from the scheduling costs of DERs and the purchase costs of electricity from the higher-level power grid, while the benefits consist of electricity sales to users and electricity sales to the higher-level power grid. The electricity price for selling to users is based on the time-of-use (TOU) tariff from [50]. To assess the impact of electricity prices on the proposed method, the electricity purchase price from the higher-level grid is set to both a fixed price (case 4) and a time-of-use price (case 5) (with the average TOU price equal to the fixed price). July 3rd is selected as a typical day for validation. The day-ahead scheduling results for case 4 and case 5 are shown in Fig. 21. The primary difference between the two methods lies in the electricity purchase power from the higher-level grid. Due to the lower TOU price between 1:00 and 6:00, case 5 results in higher electricity purchase power from the

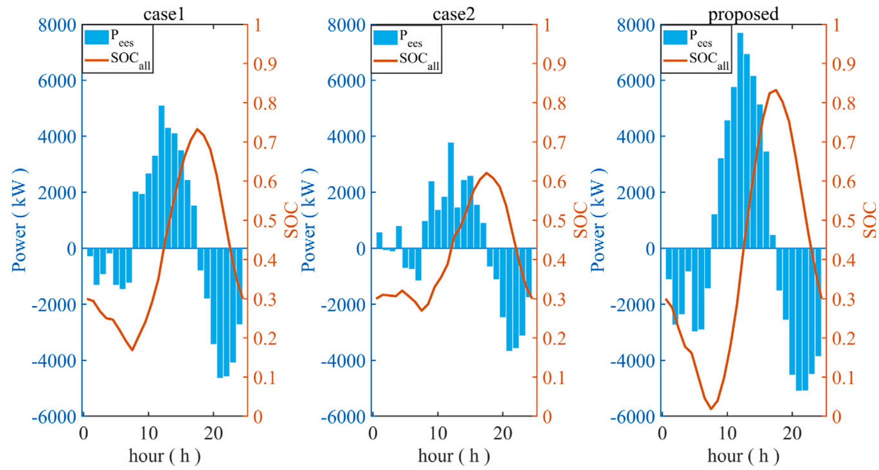


Fig. 18. The trend in SOC of EES variation corresponds with the EES output in day1.

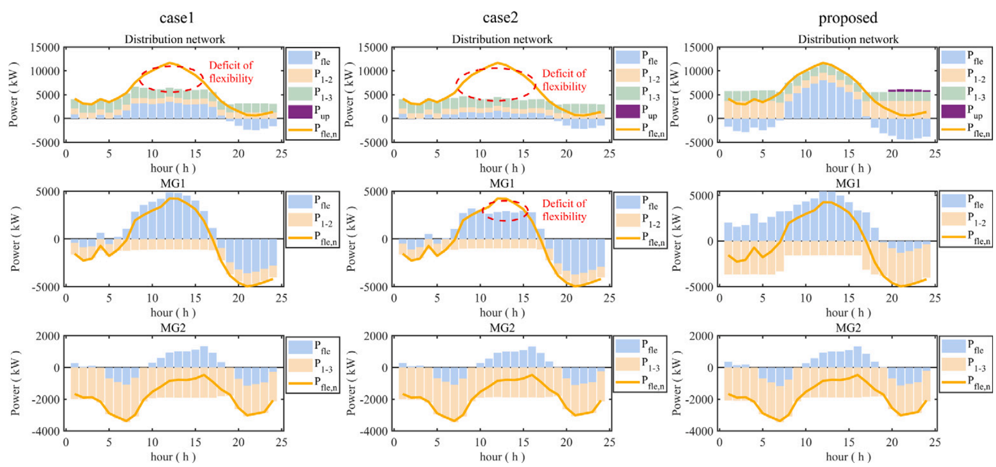


Fig. 19. Supply and demand equilibrium relationship of DS and MGs.

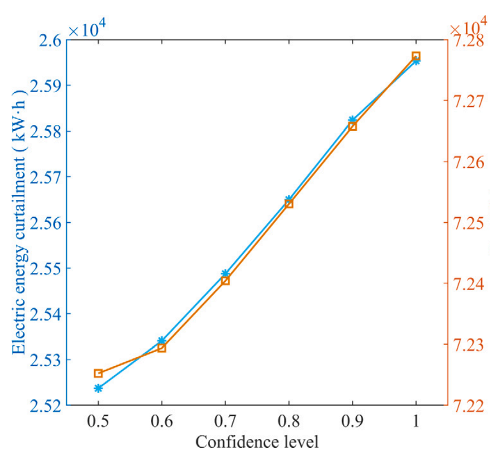


Fig. 20. The cost and electric energy curtailment of day3 at various confidence levels.

higher-level grid during this period compared to case 4, but it reduces the discharging of the energy storage system. The costs and curtailment rates for the two methods are shown in Table 2. The use of time-of-use pricing in case 5 effectively reduces the operational costs, but it leads to a higher curtailment rate. This is because case 5 involves more electricity purchase from the higher-level grid, which reduces the utilization of energy storage and increases the curtailment rate.

In addition, this section conducts a cost–benefit analysis using case 5 as an example. The Shapley value method is applied to calculate the benefit-cost allocation for each entity, requiring the solution of scheduling costs for different coalition combinations. The alliance benefit under different coalition combinations is calculated, and the Shapley values for the benefit in case 5 are presented in Table 3. Since MG1 and MG2 are

Table 2
Comparison of total system scheduling costs and power curtailment rates of case4 and case5.

	Case4	Case5
Cost ($)	9343.2	9188.4
Curtailment rates (%)	5.3961	8.4788

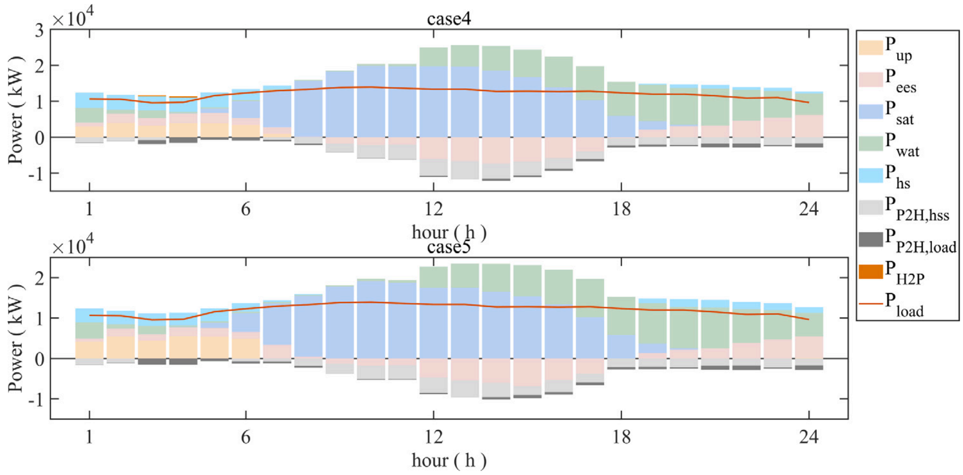


Fig. 21. Power output of case4 and case5.

Table 3
Results of the Shapley values for the benefit in case5.

Coalition scenario	No coalition	DS&MG1 or MG1&MG2 or DS&MG2 coalition	DS&MG2 or MG1&MG2 or DS&MG1 coalition	MG1&MG2&DS coalition
o(DS) − o(DS&MG1&MG2/DS)	3212.7	3531.2	3254.8	3549.8
o(MG1) − o(DS&MG1&MG2/MG1)	7577.3	7895.8	7577.3	7872.3
o(MG2) − o(DS&MG1&MG2/MG2)	7542.4	7584.5	7542.4	7561.0
$\kappa(s)$	0.33	0.17	0.17	0.33

Table 4
The benefit allocation for each entity of case6 and proposed method.

	Case6	Proposed
Benefit of DSO ($)	3212.7	3385.2
Benefit of MG1 ($)	7577.3	7728.7
Benefit of MG2 ($)	7542.4	7555.6
Total benefit ($)	18 332.4	18 669.5

not topologically connected, the alliance between MG1 and MG2 does not result in any benefit increase. In addition, compared to the no-alliance scenario, different alliance combinations all lead to an increase in benefit. Among them, MG2 contributes the most to the system's marginal benefit, followed by MG3, and lastly DS.

In order to analyze the advantages of coordinated optimal scheduling of MGs and DS, case6 is established:

case6: MGs prioritize utilizing internal flexible resources to achieve power and energy balance, accomplished by setting a high interconnection line power cost.

Based on the data in Table 3, the benefit distribution results for each participant can be derived, as shown in Table 4. Compared to case 6, the benefit allocation of proposed method for each entity is improved. The benefits of DS, MG1 and MG2 increased by $172.5, $151.4, and $13.2, respectively. This indicates that DS and MG1 contribute more to the electric power and energy balance of the entire system.

5.5. Analysis of convergence

To validate the convergence of the distributed algorithm proposed in this paper, the results for day2 were taken as an example, and a comparison is made between the computational outcomes of the traditional centralized algorithm and the distributed algorithm. In the centralized algorithm, the decision variables encompass all resources within the region, and the model's objective function and constraints align with the model presented in Section 3. Utilizing the YALMIP toolkit and invoking the GUROBI solver in the same computational environment, the centralized algorithm is solved. It is worth noting that the practical implementation of the centralized algorithm in this context is challenging, primarily due to the difficulty of controlling MG's resources from the perspective of the DS. Thus, the introduction of this algorithm is solely for contrasting the global convergence of the proposed algorithm. Fig. 22 illustrates the variation in costs with the number of iterations for both methods. It can be observed that after a finite number of iterations, the cost of the distributed algorithm gradually stabilizes and approaches that of the centralized algorithm.

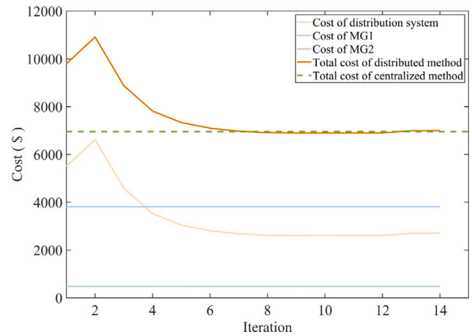


Fig. 22. Variation in costs with the number of iterations.

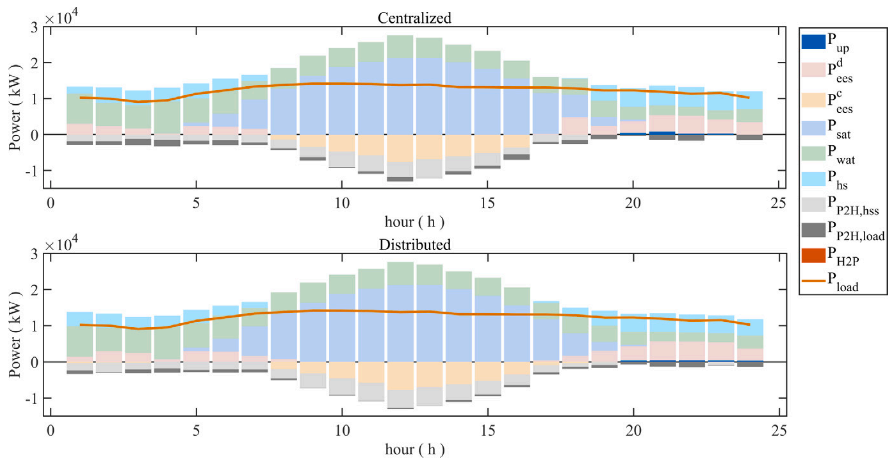


Fig. 23. Power output of centralized and distributed algorithms.

Table 5
Comparison of curtailment rates, costs, and computation time between centralized and distributed algorithms.

	Curtailment rates (%)	Costs ($)	Computation time (s)
Centralized algorithm	0.6268	6953.6	51.909
Distributed algorithm	0.6269	6998.6	116.975

Fig. 23 presents the power balance for both algorithms, showing a nearly consistent output from each resource. Furthermore, Table 5 provides a comparison of the two algorithms in terms of curtailment rates, costs, and computation time. The curtailment rates for the distributed and centralized algorithms are 0.6268% and 0.6269%, respectively, with costs of 6953.6$ and 6998.6$, showing a deviation within 0.65%. Thus, it can be concluded that the distributed algorithm achieves similar computational results to the centralized algorithm. Additionally, in terms of computation time, the distributed algorithm's serial computation time is 116.975 s. In practical applications, the distributed algorithm can employ parallel computation to enhance efficiency. If parallel computation is employed, the computation time can be reduced to 116.975/3 = 38.992 s, compared to the centralized algorithm's 51.909 s, demonstrating an improvement in computational efficiency.

Moreover, a comparison of SADMM and the traditional constant step-size ADMM algorithm in terms of iteration count is illustrated in Fig. 24. The ADMM required 30 iterations, whereas the SADMM achieved convergence in only 14 iterations. The graph depicts the variations in dual residual (r) and primal residual (s). It is evident that, at the 12th iteration, when the disparity between s and r increases, the SADMM accelerates convergence by adjusting the step size to reduce the gap between s and r.

6. Conclusion

This paper proposes a multi-time scale energy management strategy for multi-microgrid systems, taking into account multi-energy complementarity and coordinated multiple storage. The proposed method explores the coordinated integration of three different time-scale flexible resources (HS, HSS, and EES) effectively addressing power and energy balance issues at various time scales. The fuzzy theory and a variable step-size ADMM algorithm are utilized for distributed solving of the daily power balance model under uncertainty. A Shapely value based benefit allocation method between MGs and DS is proposed based on their respective contributions to the power and energy balance. Case study demonstrate that harnessing the energy regulation capabilities of HSS and HS reservoirs effectively addresses power and energy balance challenges on monthly and weekly scales, while the power regulation capabilities of EES contribute to solving daily power balance challenges. Furthermore, the adoption of a distributed energy management strategy and solving method enables power mutual aid and energy sharing between MGs and the DS, while preserving the privacy of each entity. The proposed method demonstrates its effectiveness in achieving economic efficiency and enhancing on-site consumption capacity of renewable energy for DS with MMG.

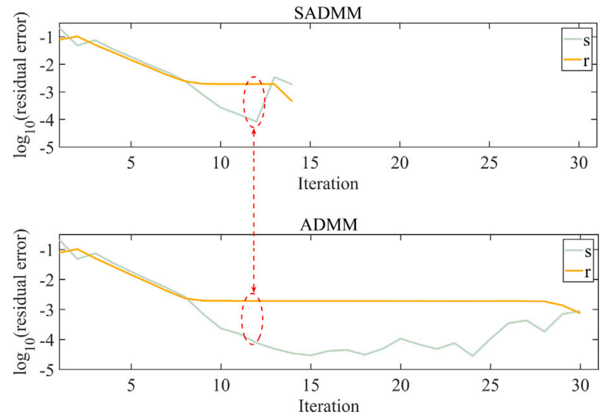


Fig. 24. The variations in dual residual and primal residual with respect to iteration for ADMM and SADMM.

Table A.1
Cost coefficients.

	c_{up} ($/kWh)	ρ_{loss} ($/kWh)	$\rho_{cle,hs}$ ($/kWh)	$\rho_{cle,pv}$ ($/kWh)	$\rho_{cle,wtg}$ ($/kWh)
Value	0.5	0.5	0.4	0.4	0.4

Table A.2
HS system parameters.

	$H_{hs,n_{hs}}$ (m)	$\eta_{hs,n_{hs}}$	$V_{hs,n_{hs},min}$ (m^3)	$V_{hs,n_{hs},max}$ (m^3)	$\sigma^m_{hs,n_{hs}}$
HS1	37.5	75%	50 000	400 000	0.3
HS2	30	78%	50 000	400 000	0.3

Table A.3
EES system parameters.

	$P^c_{es,n_{ees}}$ (MW)	$P^d_{es,n_{ees}}$ (MW)	$Q_{n_{ees}}$ (MWh)	$c^c_{ees,n_{ees}}/c^d_{ees,n_{ees}}$ ($/kWh)
EES1	7	7	14	0.0832
EES2	8	8	16	0.0798
EES3	6	6	12	0.0763
EES4	5	5	10	0.0812

However, the proposed strategy involve the exchange of data between multiple parties or systems, increasing the risk of data breaches, unauthorized access, and cyber-attacks. And the proposed method require advanced hardware or computational resources to control and communicate, which may not be feasible in all scenarios. With the advancement of privacy protection and communication control technologies, this issue can be effectively addressed. Moreover, this study has not yet explored intraday and real-time scheduling strategies. Coordinating long-duration scheduling with shorter time-scale strategies could further enhance the rationality of decision-making.

Hydrogen energy holds significant potential for ensuring the economic efficiency and flexibility of power system operations while addressing long-duration energy balance challenges. And, with the application of thermal systems in power systems, the integration of thermal systems and power systems to construct an electricity-thermal coupled integrated energy system can enhance energy utilization efficiency and facilitate the integration of renewable energy. Therefore, future research could achieve multi-energy complementarity and efficient energy utilization by integrating thermal systems with hydrogen energy and power systems more deeply.

CRediT authorship contribution statement

Peng Hao: Writing – original draft, Investigation, Methodology, Data curation. **Chunyi Huang:** Software, Writing – review & editing, Conceptualization. **Qi Wang:** Methodology, Validation. **Chengmin Wang:** Writing – review & editing, Conceptualization, Data curation. **Ning Xie:** Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables A.1–A.3.

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